EVALUATING LEVERAGED DEFI LONG–SHORT POSITIONS UNDER JUMP RISK: KILLED MOMENTS, LIQUIDATION RECOVERY AND THE LIMITS OF OPTIMAL SIZING

We develop a semi-analytic method for evaluating leveraged long–short positions held on DeFi lending protocols, whose solvency is governed by a collateral-to-liability health factor and which are liquidated when that factor falls below one. Under bivariate Kou jump–diffusion dynamics, an exponential change of measure absorbs the short-leg price and reduces every admissible killed payoff moment to a one-dimensional killed expectation of the log price ratio. Its Laplace-domain resolvent is a 3×3 linear system for every moment order, and the survival probability is its zero-order case. Bivariate tilts of the same system give the exact liquidation recovery under jump overshoot, the law of the overshoot and with it the probability of bad debt, and the expected logarithm of terminal wealth. The resulting *evaluation map* from initial health buffer to liquidation probability, payoff moments and recovery is validated against bridge-corrected Monte Carlo and is two to three orders of magnitude faster. Applied to sizing on Aave v3, the map shows that the optimal leverage is governed by a closed-form break-even drift of a few percent per year, which the data cannot estimate precisely enough to act on. A model-free replay of 37 Aave v3 pairs over 2023–2026 under the protocol's liquidation rules points the same way: modest leverage beat the unlevered position in 4 pairs, never significantly. The replay also identifies the model's main restriction: with independent jumps the calibration understates the legs' covariance, overstates spread variance and biases the break-even.

Keywords: DeFi lending; leveraged long–short; health factor; killed payoff moments; Kou double-exponential jump diffusion; first-passage problem; Laplace resolvent; jump overshoot; bad debt; Kelly criterion

JEL Classification: G11; G12; C65

1. Introduction

Crypto-asset lending protocols such as Aave and Compound enable a leveraged long–short strategy: an investor deposits one crypto asset as collateral, borrows a second, and sells it short. The position is governed by a real-time *health factor* — a borrowing-power-adjusted ratio of collateral to liability — and is liquidated when that ratio falls below one. The exposure combines leveraged relative-value risk, a path-dependent liquidation boundary, discontinuous price jumps and state-dependent survival. Its initial health buffer controls both leverage and the distance to liquidation, and evaluating that trade-off requires the distribution of a payoff killed at the liquidation boundary, at every candidate buffer.

Two objects recur: the probability that the position survives to the horizon, and the moments of its killed terminal payoff. Monte Carlo can approximate them, but barrier discretisation and sampling noise are most consequential near liquidation, and repeated simulation is costly inside a sizing search. What is missing is a fast, repeatable *evaluation map* from the initial buffer to survival probabilities, payoff moments and, once liquidation is priced, recovery — an object that any sizing rule needs and that is

independent of the rule.

We construct this map under the Kou double-exponential jump–diffusion (Kou, 2002). Two reductions make it semi-analytic. First, a unit-equity normalisation collapses the two holdings to a single initial log health factor h_0, and liquidation depends only on the log price ratio $Z = X_1 - X_2$. Second, an exponential change of measure absorbs the short-leg price factor, so every admissible payoff moment becomes a one-dimensional killed expectation of Z; because exponentials are eigenfunctions of the Kou generator, its Laplace transform solves a 3×3 linear system whose size does not grow with the moment order, and Talbot inversion (Talbot, 1979) returns the time-domain values.

The paper sits between the DeFi liquidation literature and the first-passage literature for jump diffusions. Qin et al. (2021) and Heimbach and Huang (2024) document the incentive structure and leverage amplification of DeFi liquidations empirically, without addressing ex-ante sizing. Kou and Wang (2003) derive Laplace-transform first-passage results under double-exponential jumps, and Sepp (2004) reduces barrier option pricing to a finite algebraic system by the mechanism used here; the finiteness is generic for exponential-type Lévy models (Cai, 2009; Cai and Kou, 2011; Asmussen et al., 2004). We price a barrier-killed long–short *ratio*, which the change-of-numeraire tilt brings within reach of that machinery, and use bivariate tilts of the same barrier system to obtain the joint law of the leg values at liquidation. More distantly, Jacobs et al. (1999) advocate joint optimisation of long and short legs and Dang and Forsyth (2016) study dynamic mean–variance control under jump risk; our sizing problem is a static analogue, optimising a single initial buffer against a killed-payoff distribution.

The paper makes five contributions. *(i)* It reduces the position to the scalar h_0 and constructs the objective-independent evaluation map: the liquidation probability and every admissible killed moment $\mathbb{E}[\Pi_T^k]$ from one 3×3 barrier system (Sections 2–5, Appendix A). *(ii)* It computes the exact liquidation recovery under jump overshoot from joint hitting moments under bivariate tilts, and shows that the overshoot is a creeping atom plus two exponential phases whose weights follow from three hitting moments; this gives the probability of bad debt in closed form (Section 7). *(iii)* It evaluates the expected log of terminal wealth semi-analytically, so that Kelly sizing needs no simulation (Section 7.5). *(iv)* It validates the map against an accuracy-matched, Brownian-bridge-corrected simulator and, out of sample, against realised liquidation frequencies (Section 6, Appendix C). *(v)* It applies the map to sizing. Under any of the usual objectives the optimal leverage is decided by a closed-form *break-even drift*; the data cannot estimate the drift well enough to say whether it is exceeded; and a model-free replay of 37 Aave v3 pairs shows that leverage rarely paid ex post and that the model's independent-jump restriction biases its break-evens (Section 8). Interest-rate carry enters as a drift shift and, with a Markov-modulated rate regime, through a 6×6 resolvent; both are small at a monthly horizon and are treated in Appendix D.

The framework abstracts from liquidation execution, dynamic collateral management and oracle delays, and from protocol features that rescale the effective collateral factor or cap position size (E-mode, supply/borrow caps, isolation mode). Section 2 defines the position and the evaluation map; Section 3 specifies the dynamics and tilt; Sections 4–5 derive the moments and their resolvent; Section 6 reports numerical results; Section 7 adds recovery, the overshoot law and expected log wealth; Section 8 applies the map to sizing; Section 9 concludes. The appendices give the downward-phase boundary conditions, the calibration and its full-history validation, carry and regime switching, and the proofs.

2. Leveraged Long–Short Position and Health-Factor Barrier

We consider a leveraged long–short portfolio in which an investor holds asset S_1 and owes asset S_2, with net initial equity normalized to one unit of account. The position is liquidated if collateral becomes too small relative to the liability. This section defines the barrier, the killed-payoff object studied in the paper, and the mapping from the initial health buffer to holdings. No investor objective is needed for

these definitions.

2.1. Health factor, liquidation time, and killed payoff

Let $(\Omega, \mathscr{F}, (\mathscr{F}_t)_{t\geq 0}, \mathbb{P})$ be a filtered probability space. Two strictly positive asset prices S_1, S_2 evolve as

$$S_{i,t} = S_{i,0}e^{X_{i,t}}, \qquad X_{i,0} = 0, \quad i = 1,2,$$

where X_1, X_2 are log-price processes to be specified in Section 3. The investor holds $n_1 > 0$ units of asset S_1 (long leg) and $n_2 > 0$ units of asset S_2 (short leg).

Let

$$V_{1,t} := n_1 S_{1,t}, \qquad V_{2,t} := n_2 S_{2,t}$$

denote the *collateral value* (long leg) and the *borrow value* (outstanding liability) at time t. Fix $b \in (0,1)$, the *liquidation threshold* of asset S_1 (the Aave v3 parameter of the same name; distinct from the maximum LTV at origination, which is a separate and typically lower protocol constant). Solvency requires $V_{2,t} \leq bV_{1,t}$; liquidation occurs when the borrow value exceeds adjusted collateral,

$$V_{2,t} > bV_{1,t}.$$

The *health factor* summarises this margin condition as a single ratio,

$$H_t := \frac{bV_{1,t}}{V_{2,t}} = \frac{bn_1 S_{1,t}}{n_2 S_{2,t}}, \qquad h_t := \log H_t, \tag{1}$$

where h_t is the *log health factor*. The solvency condition is $H_t \geq 1$ ($h_t \geq 0$); breaching it is equivalent to the borrow value exceeding the b-adjusted collateral. Liquidation is triggered at the first such instant:

$$\tau := \inf\{t \geq 0 : \ H_t < 1\} = \inf\{t \geq 0 : \ h_t < 0\}.$$

A larger b permits a larger borrow-to-collateral ratio $V_2/V_1 \leq b$ before liquidation, giving the position more room to absorb adverse price moves; a smaller b is a stricter protocol threshold, triggering earlier forced exit and limiting bad-debt exposure.

Since

$$h_t = \log\Big(\frac{bn_1 S_{1,0}}{n_2 S_{2,0}}\Big) + (X_{1,t} - X_{2,t}),$$

define the initial log health factor and the log-ratio process by

$$h_0 := \log\Big(\frac{bn_1 S_{1,0}}{n_2 S_{2,0}}\Big), \qquad Z_t := X_{1,t} - X_{2,t},$$

so that $h_t = h_0 + Z_t$ and the liquidation time is equivalently

$$\tau = \inf\{t \geq 0 : \ Z_t < -h_0\}.$$

The one-dimensional process Z thus carries all barrier-relevant risk: the position survives if and only if Z stays above the absorbing level $-h_0$. The parameter $h_0 = \log H_0$ is therefore the *initial log-cushion* — the log-distance of the initial health factor from the liquidation boundary $H = 1$. A larger h_0 means the position starts further from forced exit; $h_0 \to 0^+$ means the position starts almost at the boundary with negligible cushion.

The *killed terminal payoff* at horizon T is

$$\Pi_T := (n_1 S_{1,T} - n_2 S_{2,T})\,\mathbf{1}_{\{\tau > T\}}.$$

On $\{\tau > T\}$ the health factor satisfies $H_T \geq 1$, so $b n_1 S_{1,T} \geq n_2 S_{2,T}$; since $b \in (0,1)$ this forces $n_1 S_{1,T} > n_2 S_{2,T}$, meaning the position carries strictly positive terminal equity on every surviving path. No positive-part truncation is needed. The indicator $\mathbf{1}_{\{\tau>T\}}$ assigns zero to liquidated paths. This defines a *killed payoff*; it is a mathematical convention rather than a model of the investor's protocol-specific recovery. Since the initial net book value is normalised to one unit of account (Section 2.2), net profit on a surviving path is $R_T := \Pi_T - 1$, but all moments in this paper are stated in terms of Π_T directly. This paper characterises the killed moments of Π_T, the conditional distribution on $\{\tau > T\}$, and the probability of that event, with $H_t = 1$ treated as an absorbing boundary. Section 7 adds the recovery value of liquidated paths.

Remark 1 (Liquidation convention). On Aave v3 a keeper may repay up to 50% of the debt (100% below a secondary threshold) and claims a liquidation bonus of 5–15% of seized collateral; Compound is similar (Qin et al., 2021; Heimbach and Huang, 2024). Partial repayment, keeper timing, market liquidity and execution prices all affect the value retained by the borrower. We treat the barrier as absorbing and set $\Pi_T := 0$ on $\{\tau \leq T\}$ to define the killed-payoff benchmark; a partial liquidation typically restores $H > 1$ and the position continues at reduced size, so absorption models a borrower who is fully closed out. The zero is a bookkeeping convention, not a claim that recovery is worthless: investor wealth is

$$W_T = \Pi_T + R_\tau \mathbf{1}_{\{\tau \leq T\}}, \tag{2}$$

and Section 7 computes the recovery R_τ exactly, including jump overshoot.

Remark 2 (Dynamic collateral management). A practitioner who monitors the health factor could top up collateral or partially unwind the short leg to steer H_t away from the liquidation boundary. Such a strategy would replace the absorbing barrier with a reflecting or controlled boundary and the static evaluation problem with a stochastic control problem. We hold the position fixed to isolate the ex-ante killed-moment calculation; the assumption should not be interpreted as a claim that dynamic intervention is infeasible or suboptimal. During a flash crash — the jump-driven event that dominates liquidation — a position may be liquidated within one or two blocks, before a remedial transaction confirms, so the static model is accurate for stress episodes short relative to a monitoring cycle.

Remark 3 (Oracle prices). Aave liquidates against Chainlink prices that update on a heartbeat or deviation threshold. Discrete monitoring means a dip that recovers between updates does not liquidate, so the continuous barrier *overstates* liquidation frequency; but a stale oracle that gaps down liquidates at the gapped level, so the continuous model *understates* the recovery loss. Calibration prices are reserve-level oracle-implied values (TVL divided by supplied balance).

Thus killed payoff and investor terminal wealth are distinct objects. The former is the tractable object evaluated below; recovery and intervention overlays may change the latter and, consequently, any sizing decision built from it.

Using the definition of H_T, one obtains the following factorization.

Lemma 2.1 (Payoff factorization)**.**

$$\Pi_T = \frac{n_2}{b} S_{2,0} e^{X_{2,T}} \left(e^{h_0 + Z_T} - b\right) \mathbf{1}_{\{\tau>T\}}. \tag{3}$$

The factor $(e^{h_0+Z_T} - b)$ *is strictly positive on* $\{\tau > T\}$ *because* $H_T = e^{h_0+Z_T} \geq 1 > b$*, consistent with the definition of* Π_T*.*

This separates the payoff into a numeraire factor driven by S_2, and a barrier-killed option on the log-ratio Z with moneyness controlled by h_0.

2.2. Health-buffer parameterization and feasible set

The two coin holdings (n_1, n_2) represent two degrees of freedom. A normalization reduces this to a single scalar, and the barrier structure identifies h_0 as the natural evaluation parameter. This reduction does not itself select a portfolio.

We impose the unit-initial-value constraint

$$n_1 S_{1,0} - n_2 S_{2,0} = 1,$$

which fixes the net book value at inception. Setting $x := n_2 S_{2,0}$, one has $n_1 S_{1,0} = 1 + x$ and $e^{h_0} = b(1 + x)/x$. Solving for x,

$$x = \frac{b}{e^{h_0} - b},$$

which yields the holding map

$$n_2(h_0) = \frac{b}{S_{2,0}(e^{h_0} - b)}, \qquad n_1(h_0) = \frac{e^{h_0}}{S_{1,0}(e^{h_0} - b)}. \tag{4}$$

Positivity of both holdings requires $e^{h_0} > b$, i.e.

$$h_0 > \ln b.$$

This algebraic condition is necessary but not sufficient for a feasible DeFi position. Initial solvency additionally requires $H_0 = e^{h_0} > 1$, hence $h_0 > 0$. If the protocol imposes a maximum loan-to-value ratio $\mathrm{LTV}_{\max} \in (0, b)$ at origination, then

$$\frac{V_{2,0}}{V_{1,0}} = \frac{b}{H_0} \le \mathrm{LTV}_{\max} \quad \Longleftrightarrow \quad h_0 \ge h_0^{\min} := \log\left(\frac{b}{\mathrm{LTV}_{\max}}\right) > 0.$$

Accordingly, define the feasible set

$$\mathscr{H}(b, \mathrm{LTV}_{\max}) := \begin{cases} [\log(b/\mathrm{LTV}_{\max}), \infty), & \text{when an origination LTV } \mathrm{LTV}_{\max} \in (0, b) \text{ applies,} \\ (0, \infty), & \text{otherwise.} \end{cases} \tag{5}$$

The lower endpoint is included in the first case because the protocol permits origination at its maximum LTV. The point $h_0 = 0$ is excluded in the second case because it starts the position at the liquidation boundary. The initial gross leverage — long-leg value per unit of equity — follows directly from (4):

$$L_0 := n_1(h_0)\, S_{1,0} = \frac{e^{h_0}}{e^{h_0} - b}. \tag{6}$$

L_0 is strictly decreasing in h_0. Algebraically, $L_0 \uparrow \infty$ as $h_0 \downarrow \ln b$, but this limit is infeasible because $\ln b < 0$ and the position would already be liquidatable at inception. Over the feasible set (5), maximum leverage is attained at the smallest permitted health buffer; as $h_0 \to \infty$, leverage $L_0 \downarrow 1$ (the short leg vanishes and the position becomes unlevered). Equivalently, the borrow-to-equity ratio is $L_0 - 1 = b/(e^{h_0} - b)$, which also collapses the initial health factor back to $H_0 = e^{h_0}$ via $H_0 = b\, L_0/(L_0 - 1)$. Choosing h_0 is therefore equivalent to choosing the initial leverage.

For every feasible h_0, the objective-independent model output is the evaluation map

$$h_0 \longmapsto \left(p_{\mathrm{liq}}(T, h_0), M_1(T, h_0), \ldots, M_K(T, h_0), \frac{M_1(T, h_0)}{p_{\mathrm{surv}}(T, h_0)}, \ldots, \frac{M_K(T, h_0)}{p_{\mathrm{surv}}(T, h_0)} \right), \tag{7}$$

where $M_k(T,h_0)=\mathbb{E}[\Pi_T^k]$ and K is any fixed admissible integer order. Sections 4–5 construct this map, and Section 7 extends it with the moments of W_T, the overshoot law and $\mathbb{E}[\log W_T]$. A decision maker subsequently applies a liquidation limit or a criterion to it (Section 8). For mean–variance we use *unconditional* terminal wealth,

$$\Phi(h_0) := \mathbb{E}[W_T] - \tfrac{\alpha}{2}\operatorname{Var}(W_T), \qquad \alpha \geq 0. \tag{8}$$

The survival-conditional moments $\mathbb{E}[\Pi_T^k \mid \tau > T] = \mathbb{E}[\Pi_T^k]/p_{\text{surv}}$ are useful diagnostics but should not be maximised: conditioning keeps only favourable paths, so the conditional mean rises as the buffer thins toward the barrier (Table 3) and rewards leverage for a survivorship artefact.

3. Bivariate Kou Dynamics and Exponential Tilting

3.1. Bivariate Kou model

We assume under $\mathbb{P}$ that each log-price satisfies the SDE

$$\mathrm{d}X_{i,t} = \mu_i^X\,\mathrm{d}t + \sigma_i\,\mathrm{d}B_{i,t} + \mathrm{d}\left(\sum_{n=1}^{N_{i,t}} J_i^{(n)}\right), \qquad i=1,2, \tag{9}$$

where B_1, B_2 are correlated standard Brownian motions with $\mathrm{d}\langle B_1,B_2\rangle_t = \rho\,\mathrm{d}t$, N_i is a Poisson process with intensity λ_i independent of (B_1,B_2), and the jump sizes $J_i^{(n)}$ are i.i.d. with the Kou double-exponential law. The jump density, characteristic function, and moment generating function are

$$f_i(j) = \frac{p_i}{\eta_{i,+}} e^{-j/\eta_{i,+}}\mathbf{1}_{\{j>0\}} + \frac{1-p_i}{\eta_{i,-}} e^{j/\eta_{i,-}}\mathbf{1}_{\{j<0\}}, \qquad \varphi_{J_i}(u) = \frac{p_i}{1-iu\eta_{i,+}} + \frac{1-p_i}{1+iu\eta_{i,-}}, \tag{10}$$

$$M_i(s) = \frac{p_i}{1-s\eta_{i,+}} + \frac{1-p_i}{1+s\eta_{i,-}}, \qquad s \in \left(-\tfrac{1}{\eta_{i,-}}, \tfrac{1}{\eta_{i,+}}\right). \tag{11}$$

Here $\eta_{i,+}$ and $\eta_{i,-}$ are the *means* of the positive and negative jump sizes (not exponential rates); the corresponding rates are $1/\eta_{i,+}$ and $1/\eta_{i,-}$. The constraint $\eta_{i,+}<1$ is required for the price-jump compensator χ_i defined below to be finite.

For each asset, we use a single free drift parameter, the *expected-price growth exponent* μ_i, defined by

$$\mathbb{E}_{\mathbb{P}}[S_{i,t}/S_{i,0}] = e^{\mu_i t}. \tag{12}$$

This is not the mean log-return rate. Define the price-jump compensator

$$\chi_i := M_i(1) - 1 = \mathbb{E}[e^{J_i}-1] = \frac{p_i}{1-\eta_{i,+}} + \frac{1-p_i}{1+\eta_{i,-}} - 1. \tag{13}$$

Since $\mathbb{E}_{\mathbb{P}}[e^{X_{i,t}}] = \exp(t[\mu_i^X + \tfrac{1}{2}\sigma_i^2 + \lambda_i\chi_i])$, matching (12) gives the drift coefficient in the log-price SDE,

$$\mu_i^X := \mu_i - \tfrac{1}{2}\sigma_i^2 - \lambda_i\chi_i. \tag{14}$$

Thus μ_i^X is a derived shorthand used in the characteristic exponent and subsequent calculations, not an additional model parameter.

The bivariate process (X_1,X_2) is a Lévy process with joint characteristic exponent

$$\begin{aligned}\Psi(u,v) := {} & i(\mu_1^X u + \mu_2^X v) - \tfrac{1}{2}\left(\sigma_1^2u^2 + \sigma_2^2v^2 + 2\rho\sigma_1\sigma_2uv\right) \\ & + \lambda_1\left(\varphi_{J_1}(u)-1\right) + \lambda_2\left(\varphi_{J_2}(v)-1\right),\end{aligned} \tag{15}$$

so that $\mathbb{E}_{\mathbb{P}}[e^{i(uX_{1,t}+vX_{2,t})}] = e^{t\Psi(u,v)}$.

3.2. Tilted dynamics

For each integer $k \geq 0$, define the k-tilted measure $\mathbb{P}^{(k)}$ by

$$\left.\frac{\mathrm{d}\mathbb{P}^{(k)}}{\mathrm{d}\mathbb{P}}\right|_{\mathscr{F}_t} = \frac{e^{kX_{2,t}}}{\mathbb{E}_{\mathbb{P}}[e^{kX_{2,t}}]} = \exp(kX_{2,t} - t\,\Psi(0,-ik))\,, \qquad k\eta_{2,+} < 1.$$

Lemma 3.1 (Lévy exponent under $\mathbb{P}^{(k)}$)**.** *Under $\mathbb{P}^{(k)}$, the log-ratio $Z = X_1 - X_2$ is again a Lévy process with exponent*

$$\psi_Z^{(k)}(s) = \mu_Z^{(k)} s + \tfrac{1}{2}\sigma_Z^2 s^2 + \lambda_1\big(M_1(s) - 1\big) + \lambda_2\big(M_2(k-s) - M_2(k)\big), \tag{16}$$

where

$$\sigma_Z^2 := \sigma_1^2 + \sigma_2^2 - 2\rho\sigma_1\sigma_2 \geq 0, \qquad \mu_Z^{(k)} := (\mu_1^X - \mu_2^X) - k(\sigma_2^2 - \rho\sigma_1\sigma_2).$$

In particular, the generator satisfies

$$\mathscr{L}^{(k)} e^{sz} = \psi_Z^{(k)}(s)\, e^{sz}. \tag{17}$$

▷ *Proof, p. 34*

This eigenrelation is the key identity used in Section 5 to turn the Laplace-domain problem into a finite exponential representation.

Remark 4 (Superscript (k) convention). A superscript (k) indicates evaluation under $\mathbb{P}^{(k)}$. The untilted case $k = 0$ recovers the original dynamics of Z under $\mathbb{P}$. For $k \geq 1$, the k-th moment $\mathbb{E}[\Pi_T^k]$ is computed under $\mathbb{P}^{(k)}$. The constraint $k\eta_{2,+} < 1$ makes the tilted measure well-defined, while the payoff forcing additionally requires $k\eta_{1,+} < 1$ so that the long-leg exponential moments $M_1(j)$ are finite for $j = 0, \ldots, k$. Thus a payoff-moment order is *admissible* when

$$k\eta_{1,+} < 1, \qquad k\eta_{2,+} < 1. \tag{18}$$

Throughout, k is an integer. There is no restriction to $k = 1$ or $k = 2$: every fixed positive integer satisfying (18) is covered.

For the Kou structure, the tilt transforms the jump rates of Z into the k-dependent phase rates

$$r_{1,+}^{(k)} := \frac{1}{\eta_{1,+}}, \quad r_{1,-}^{(k)} := \frac{1}{\eta_{1,-}}, \quad r_{2,+}^{(k)} := \frac{1}{\eta_{2,-}} + k, \quad r_{2,-}^{(k)} := \frac{1}{\eta_{2,+}} - k.$$

Because $Z = X_1 - X_2$, positive jumps of X_2 appear as downward jumps of Z: the tilt by e^{kX_2} changes the downward phase rate from $1/\eta_{2,+}$ to $1/\eta_{2,+} - k$, which is positive precisely when $k\eta_{2,+} < 1$. The rates $r_{1,-}^{(k)}$ and $r_{2,-}^{(k)}$ enter the finite-dimensional system in Section 5. The associated phase operators and their truncation at the liquidation boundary are developed separately in Appendix A.

Table 1 collects the quantities introduced in this subsection.

4. Reduction of Killed Payoff Moments

Fix any integer $k \geq 1$ satisfying the two moment-domain conditions (18). Computing $\mathbb{E}_{\mathbb{P}}[\Pi_T^k]$ directly under $\mathbb{P}$ is intractable: the payoff couples the joint process (X_1, X_2) to the barrier event $\{\tau > T\}$ in a way that is neither separable nor analytically tractable. An exponential change of measure resolves this by tilting the path measure by $e^{kX_{2,T}}$, which absorbs the short-leg factor into the density and leaves a purely one-dimensional killed expectation for the log-ratio $Z = X_1 - X_2$ under the k-tilted measure $\mathbb{P}^{(k)}$.

Symbol	Definition	Role
$\mathbb{P}^{(k)}$	$d\mathbb{P}^{(k)}/d\mathbb{P} \propto e^{kX_{2,t}}$	Measure under which moment reduction is carried out
k	Non-negative integer; (18) for $k \geq 1$	Tilting order; $k \geq 1$ for moments, $k = 0$ for survival probability
$\psi_Z^{(k)}(s)$	Lévy exponent of Z under $\mathbb{P}^{(k)}$	Characteristic root equation $\psi_Z^{(k)}(\gamma) = q$ governs the resolvent
σ_Z^2	$\sigma_1^2 + \sigma_2^2 - 2\rho\sigma_1\sigma_2$	Variance of the Brownian part of Z; appears in $\psi_Z^{(k)}$
$\mu_Z^{(k)}$	$(\mu_1^X - \mu_2^X) - k(\sigma_2^2 - \rho\sigma_1\sigma_2)$	Drift of Z under $\mathbb{P}^{(k)}$; appears in $\psi_Z^{(k)}$
$\mathscr{L}^{(k)}$	Generator of Z under $\mathbb{P}^{(k)}$	$\mathscr{L}^{(k)} e^{sz} = \psi_Z^{(k)}(s) e^{sz}$; killing is imposed by zero extension in Section 5
$r_{j,\pm}^{(k)}$	Phase rates (see above)	Exponential rates of the jump components of Z under $\mathbb{P}^{(k)}$

Table 1: Tilted-dynamics quantities. Subscript $j = 1$ refers to the contribution from X_1 jumps; $j = 2$ to that from X_2 jumps (sign-reversed because $Z = X_1 - X_2$).

Since Z remains a Lévy process under $\mathbb{P}^{(k)}$, no information about the joint dynamics is lost: the tilted characteristic exponent encodes the full marginal law of Z. The k-th moment then factorises as the known scalar $e^{T\Psi(0,-ik)} = \mathbb{E}_{\mathbb{P}}[e^{kX_{2,T}}]$ times the barrier-killed expectation $m_k(T;h_0)$ under $\mathbb{P}^{(k)}$, which is computed via the Laplace–resolvent of Section 5.

Define

$$g_k(z;h_0) := (e^{h_0+z} - b)^k.$$

Since the survival event $\{\tau > T\}$ implies $Z_T > -h_0$, we have $e^{h_0+Z_T} > 1 > b$ on $\{\tau > T\}$, so $g_k(Z_T;h_0) = (e^{h_0+Z_T} - b)^k > 0$ and is a polynomial in e^{Z_T}. From (3),

$$\Pi_T^k = \left(\frac{n_2}{b}\right)^k S_{2,0}^k e^{kX_{2,T}}\, g_k(Z_T;h_0)\, \mathbf{1}_{\{\tau>T\}}.$$

$$\mathbb{E}_{\mathbb{P}}[\Pi_T^k] = \left(\frac{n_2}{b}\right)^k S_{2,0}^k\, e^{T\Psi(0,-ik)}\, \mathbb{E}_{\mathbb{P}^{(k)}}\Big[g_k(Z_T;h_0)\, \mathbf{1}_{\{\tau>T\}}\Big]. \tag{19}$$

Proposition 4.1 (Every admissible integer-order moment)**.** *Let $k \geq 1$ be an integer satisfying (18). Define*

$$m_k(T;h_0) := \mathbb{E}_{\mathbb{P}^{(k)}}\Big[g_k(Z_T;h_0)\, \mathbf{1}_{\{\tau>T\}}\Big].$$

Then

$$\mathbb{E}_{\mathbb{P}}[\Pi_T^k] = \frac{e^{T\Psi(0,-ik)}}{(e^{h_0} - b)^k}\, m_k(T;h_0),$$

where $m_k(T;h_0)$ is computed via the Laplace–resolvent method of Section 5. Consequently, the method computes $\mathbb{E}[\Pi_T^k]$ for every integer

$$k \in \mathscr{K} := \{n \in \mathbb{N} :\ n\eta_{1,+} < 1 \text{ and } n\eta_{2,+} < 1\}.$$

Let $p_{\text{surv}}(h_0) := \mathbb{P}(\tau > T)$ and $p_{\text{liq}}(h_0) := 1 - p_{\text{surv}}(h_0)$. Since Π_T already contains the survival indicator $\mathbf{1}_{\{\tau>T\}}$, the conditional k-th moment on surviving paths is

$$\mathbb{E}[\Pi_T^k \mid \tau > T] = \frac{\mathbb{E}[\Pi_T^k]}{p_{\text{surv}}(h_0)}, \qquad k \geq 1, \quad k\eta_{1,+} < 1, \quad k\eta_{2,+} < 1.$$

In particular,

$$\mathbb{E}[\Pi_T \mid \tau > T] = \frac{\mathbb{E}[\Pi_T]}{p_{\text{surv}}(h_0)}, \qquad \text{Var}(\Pi_T \mid \tau > T) = \frac{\mathbb{E}[\Pi_T^2]}{p_{\text{surv}}(h_0)} - \left(\frac{\mathbb{E}[\Pi_T]}{p_{\text{surv}}(h_0)}\right)^2.$$

The same killed moments also give the unconditional variance of the zero-after-liquidation benchmark,

$$\mathrm{Var}(\Pi_T) = \mathbb{E}[\Pi_T^2] - \mathbb{E}[\Pi_T]^2.$$

Thus survival, killed moments, unconditional moments, and conditional surviving-path moments are all outputs of the same construction. No preference parameter enters this reduction; Section 8 later combines these outputs under several decision rules.

Remark 5 (No-shorting limit). As $h_0 \to \infty$ the holding map gives $n_2 S_{2,0} \to 0$ and $n_1 S_{1,0} \to 1$, so liquidation becomes impossible and the portfolio converges to a unit long position in asset 1. For every admissible integer $k \geq 1$,

$$\lim_{h_0 \to \infty} \mathbb{E}[\Pi_T^k] = \mathbb{E}[e^{kX_{1,T}}] = e^{T\Psi(-ik,0)},$$

which provides a useful diagnostic: the formula for $\mathbb{E}[\Pi_T^k]$ must satisfy this limit exactly. The $k = 1$ case gives $e^{\mu_1 T}$ by the drift normalisation (14).

The remainder of the analysis is devoted to computing $m_k(T;h_0)$ for every admissible integer $k \geq 1$ and $p_{\mathrm{surv}}(h_0)$ via the Laplace–resolvent construction.

5. Laplace Transforms of the Killed Moments

This section computes the time-Laplace transform of the reduced killed moment $m_k(T;h_0)$ from Proposition 4.1. We first express that target as a killed resolvent, then evaluate it through a fixed-dimensional linear system. The lower-boundary calculation specific to downward jumps is isolated in Appendix A.

5.1. Target Transform and Resolvent

Fix an integer $k \geq 0$ satisfying (18), and define

$$G_k(x) := (e^x - b)^k = \sum_{j=0}^{k} d_{k,j} e^{jx}, \qquad d_{k,j} := \binom{k}{j} (-b)^{k-j}. \tag{20}$$

For a starting distance $x > 0$ above the translated barrier, define the killed moment function directly by

$$\widetilde{U}_k(t,x) := \mathbb{E}_{\mathbb{P}^{(k)}}\left[G_k(x + Z_t) \mathbf{1}_{\{\inf_{0 \leq s \leq t}(x+Z_s) > 0\}} \right].$$

Because $\sigma_Z > 0$, zero is regular for the negative half-line, so this survival event agrees almost surely with the strict-breach convention $x + Z_s < 0$ used in Section 2. Then $m_k(t;h_0) = \widetilde{U}_k(t,h_0)$ for $k \geq 1$. For uniform notation, set $p_{\mathrm{surv}}(t,h_0) := \mathbb{P}(\tau > t)$ and $m_0(t;h_0) := p_{\mathrm{surv}}(t,h_0)$. Since $G_0 \equiv 1$, the same identity $m_k(t;h_0) = \widetilde{U}_k(t,h_0)$ then holds for every $k \geq 0$ considered below.

For q to the right of the time-Laplace abscissa of convergence, define

$$\widehat{U}_k(q,x) := \int_0^\infty e^{-qt} \widetilde{U}_k(t,x)\, \mathrm{d}t, \qquad \widehat{m}_k(q;h_0) := \int_0^\infty e^{-qt} m_k(t;h_0)\, \mathrm{d}t = \widehat{U}_k(q,h_0). \tag{21}$$

At $k = 0$, (21) is the Laplace transform of the survival probability.

We regard $\widetilde{U}_k(t,\cdot)$ as a function on $\mathbb{R}$ by setting it equal to zero on $(-\infty, 0]$. With killing encoded by this convention,

$$\partial_t \widetilde{U}_k(t,x) = \mathscr{L}^{(k)}\left[\widetilde{U}_k(t,\cdot) \right](x), \qquad x > 0, \tag{22}$$

with initial condition $\widetilde{U}_k(0,x) = G_k(x)$ for $x > 0$. Integrating (22) by parts in t gives

$$q\widehat{U}_k(q,x) - G_k(x) = \mathscr{L}^{(k)}\left[\widehat{U}_k(q,\cdot)\right](x).$$

Equivalently, under the zero-extension convention, the target solves

$$(q - \mathscr{L}^{(k)})\widehat{U}_k(q,x) = G_k(x), \qquad x > 0, \qquad \widehat{U}_k(q,x) = 0 \quad (x \le 0). \tag{23}$$

Appendix A verifies how the zero extension acts on the jump terms and derives the boundary equations used below.

5.2. Time-Laplace Transform and Finite-Dimensional System

We solve (23) under the following regularity assumption.

Assumption 5.1 (Generic six-root regime)**.** Fix an integer $k \ge 0$ satisfying (18). Let real q lie strictly to the right of the relevant time-Laplace abscissa of convergence. Assume:

(i) $\sigma_Z > 0$;

(ii) the characteristic equation $\psi_Z^{(k)}(\gamma) = q$ has six simple roots $\gamma_1^{(k)}(q), \dots, \gamma_6^{(k)}(q)$, ordered so that

$$\mathrm{Re}(\gamma_{1,2,3}^{(k)}) > 0, \qquad \mathrm{Re}(\gamma_{4,5,6}^{(k)}) < 0;$$

(iii) $q \neq \psi_Z^{(k)}(j)$ for $j = 0, \dots, k$.

For real $q > 0$ with distinct poles ($r_{1,+}^{(k)} \neq r_{2,+}^{(k)}$ and $r_{1,-}^{(k)} \neq r_{2,-}^{(k)}$) the three-and-three split is automatic. Each Kou term of $\psi_Z^{(k)}$ changes sign through its pole, tending to $+\infty$ on the side facing the origin and to $-\infty$ on the other; $\psi_Z^{(k)}(0) - q = -q < 0$; and the diffusion term sends $\psi_Z^{(k)} - q$ to $+\infty$ as $|\gamma| \to \infty$. By the intermediate value theorem there is a real root in each of the six intervals cut out by the origin and the poles $\{r_{1,+}^{(k)}, r_{2,+}^{(k)}, -r_{1,-}^{(k)}, -r_{2,-}^{(k)}\}$, three on each side, and since clearing denominators gives a polynomial of degree six these are all the roots and they are simple (Kou and Wang, 2003; Sepp, 2004; Cai and Kou, 2011). The assumption has content for the complex q at which the Talbot contour evaluates the transform, part of which lies in $\mathrm{Re}\, q < 0$. There the roots are continued analytically from the real case, and the implementation classifies them by the rank of their real parts rather than by sign. At the headline calibration the sign split holds at every Talbot node, and node counts from $M = 16$ to 48 agree to 10^{-8}.

The forcing expansion (20) and generator eigenrelation (17) motivate the interior ansatz

$$\widehat{U}_k(q,x) = \sum_{j=0}^{k} \frac{d_{k,j}}{q - \psi_Z^{(k)}(j)} e^{jx} + \sum_{m=1}^{3} B_m^{(k)}(q) e^{\gamma_{m+3}^{(k)}(q)x}, \qquad x > 0. \tag{24}$$

The exponential-root structure of (24) closely follows the Laplace-transform construction of Sepp (2004) for barrier options under double-exponential jump diffusion, adapted here to a forced resolvent for killed moments. In this adaptation, the first sum is the particular response to the forcing G_k, while the second is a homogeneous barrier correction. As the starting distance $x \to +\infty$, the effect of the lower barrier must vanish and the killed resolvent must approach its unrestricted counterpart. A term $e^{\gamma x}$ with $\mathrm{Re}(\gamma) > 0$ would instead diverge, so the three positive-real-part roots are inadmissible. The ansatz therefore retains only the three roots with negative real part, whose exponentials decay. Their coefficients $B_m^{(k)}(q)$ remain to be determined. The following proposition states the desired transform; Appendix A supplies the downward-phase boundary calculation behind its three matrix rows.

Proposition 5.1 (Laplace-transformed killed moments)**.** *Under Assumption 5.1 and the nondegeneracy condition*

$$r_{1,-}^{(k)} \neq r_{2,-}^{(k)}, \tag{25}$$

the coefficients $\mathbf{B}^{(k)}(q) := (B_1^{(k)}, B_2^{(k)}, B_3^{(k)})^\top$ *are the unique solution of*

$$\mathbf{M}_{\mathrm{bar}}^{(k)}(q)\,\mathbf{B}^{(k)}(q) = -\mathbf{b}^{(k)}(q), \tag{26}$$

where

$$\mathbf{M}_{\mathrm{bar}}^{(k)}(q) = \begin{pmatrix} 1 & 1 & 1 \\ \dfrac{r_{1,-}^{(k)}}{r_{1,-}^{(k)} + \gamma_4^{(k)}(q)} & \dfrac{r_{1,-}^{(k)}}{r_{1,-}^{(k)} + \gamma_5^{(k)}(q)} & \dfrac{r_{1,-}^{(k)}}{r_{1,-}^{(k)} + \gamma_6^{(k)}(q)} \\ \dfrac{r_{2,-}^{(k)}}{r_{2,-}^{(k)} + \gamma_4^{(k)}(q)} & \dfrac{r_{2,-}^{(k)}}{r_{2,-}^{(k)} + \gamma_5^{(k)}(q)} & \dfrac{r_{2,-}^{(k)}}{r_{2,-}^{(k)} + \gamma_6^{(k)}(q)} \end{pmatrix} \tag{27}$$

and

$$\mathbf{b}^{(k)}(q) = \begin{pmatrix} \displaystyle\sum_{j=0}^{k} \frac{d_{k,j}}{q - \psi_Z^{(k)}(j)} \\ \displaystyle\sum_{j=0}^{k} \frac{d_{k,j}}{q - \psi_Z^{(k)}(j)} \dfrac{r_{1,-}^{(k)}}{r_{1,-}^{(k)} + j} \\ \displaystyle\sum_{j=0}^{k} \frac{d_{k,j}}{q - \psi_Z^{(k)}(j)} \dfrac{r_{2,-}^{(k)}}{r_{2,-}^{(k)} + j} \end{pmatrix}. \tag{28}$$

By Lemma 5.2, the matrix (27) is invertible, so

$$\mathbf{B}^{(k)}(q) = -\big(\mathbf{M}_{\mathrm{bar}}^{(k)}(q)\big)^{-1}\mathbf{b}^{(k)}(q).$$

The Laplace-transformed killed moment is therefore

$$\boxed{\widehat{m}_k(q;h_0) = \widehat{U}_k(q,h_0) = \sum_{j=0}^{k} \frac{d_{k,j}}{q - \psi_Z^{(k)}(j)} e^{jh_0} + \sum_{m=1}^{3} B_m^{(k)}(q) e^{\gamma_{m+3}^{(k)}(q)h_0}.} \tag{29}$$

▷ *Proof, p. 35*

Lemma 5.2 (Invertibility of $\mathbf{M}_{\mathrm{bar}}^{(k)}(q)$)**.** *Under Assumption 5.1 and condition (25), the barrier matrix* $\mathbf{M}_{\mathrm{bar}}^{(k)}(q)$ *in (27) is invertible.*

▷ *Proof, p. 35*

The lemma ensures that the absorption condition and the two downward-phase cancellation conditions determine the three decaying-mode coefficients uniquely.

The generic system (26) assumes distinct downward rates. If the two rates coincide, they form one phase and the reduced construction is 2×2; the details are given in Appendix A, Remark 7.

5.3. Moment Recovery and Survival

Numerical Laplace inversion of (29) recovers the time-domain killed moment,

$$m_k(T;h_0) = \mathscr{L}_{q\to T}^{-1}[\widehat{m}_k(q;h_0)]\,. \tag{30}$$

The survival probability is the same construction at $k=0$. Since $\psi_Z^{(0)}(0)=0$, one has $P_0(q,x)=1/q$ and, when $\eta_{1,-}\neq\eta_{2,+}$,

$$\widehat{p}_{\mathrm{surv}}(q;h_0)=\widehat{U}_0(q,h_0)=\frac{1}{q}+\sum_{m=1}^{3}B_m^{(0)}(q)e^{\gamma_{m+3}^{(0)}(q)h_0},\qquad \mathbf{M}_{\mathrm{bar}}^{(0)}(q)\mathbf{B}^{(0)}(q)=-\begin{pmatrix}1/q\\1/q\\1/q\end{pmatrix}. \tag{31}$$

Hence

$$p_{\mathrm{surv}}(T,h_0)=\mathscr{L}^{-1}_{q\to T}[\widehat{p}_{\mathrm{surv}}(q;h_0)],\qquad p_{\mathrm{liq}}(T,h_0)=1-p_{\mathrm{surv}}(T,h_0).$$

If the two untilted downward rates coincide, the $k=0$ instance of Appendix A, Remark 7, gives the corresponding 2×2 system.

6. Numerical Results

We validate the resolvent against Monte Carlo (MC) for $k=0,\dots,4$ — an illustration, not a limit: the same code handles every $k\in\mathscr{K}$ — and check the no-shorting limit. Inversion uses the Talbot contour (Talbot, 1979) with $M=32$ nodes.

The diffusion, jump and correlation parameters (Table 2) are calibrated by penalised ECF matching (Appendix B) to 840 four-hour WBTC (asset 1) and WETH (asset 2) log-returns on Aave v3 Ethereum, after subtracting an exponentially weighted mean. The liquidation threshold $b=0.78$ and maximum origination LTV 0.73 give a minimum feasible buffer $h_0^{\min}=\log(0.78/0.73)\approx 0.066$. WBTC and WETH have distinct downward phase rates, so (25) holds for every admissible k.

The drift is kept outside the ECF fit. High-frequency sampling sharpens volatility estimates much more than drift estimates (Merton, 1980), and on de-meaned returns a free drift in the characteristic-function phase confounds location with jump shape: the fitted drift parameters are normalizers, whose difference (-0.86) even has the opposite sign to the window's realised spread drift. The benchmark therefore states its drift explicitly. It uses the window's sample drifts in the price-growth convention of (12), including the Aave supply and borrow rates, $\mu_1=-0.314$ and $\mu_2=-0.417$, so that the spread drift is $\nu=\mu_1-\mu_2=+0.103$ per year ($+0.128$ in price growth, -0.025 in net carry). Both legs fell over the window. Any other stated view can be substituted; Section 8 shows how much the sizing answer depends on it.

For $h_0\in\{0.07,0.10,0.25,0.50,0.75,1.00,1.50,2.00\}$ Table 3 reports p_{surv} and the survival-conditional mean, variance, skewness and excess kurtosis of Π_T, computed from $\mu_k:=\mathbb{E}[\Pi_T^k\mid\tau>T]$ in the usual way, together with the $h_0\to\infty$ limits $\mu_k\to\exp(T\Psi(-ik,0))$ of Remark 5.

The conditional moments agree with MC to within 0.004 (mean), 0.0004 (variance), 0.016 (skewness) and 0.07 (excess kurtosis) at every tabulated buffer. In this run the Laplace method took 0.3–0.6 s per buffer for all five inversions against 104–117 s for MC; timings are machine- and load-dependent, so only the order of magnitude is meaningful. The only systematic discrepancy is in survival: MC exceeds Laplace by about one point near $h_0^{\min}$ and by nothing for $h_0\geq 0.5$. Section 6.1 shows this is MC discretisation, not analytic error.

6.1. An accuracy-matched Monte Carlo benchmark

The Euler–Poisson scheme checks the barrier only at step ends, so it misses intra-step crossings and *overstates* survival. An accuracy-matched simulator draws the diffusion on a grid and adds the Brownian-bridge probability of an intra-step crossing analytically (Glasserman, 2004), with jumps applied and checked at step ends. At $h_0=0.07$ (Table 4) the bridge-corrected estimate converges to the Laplace

Parameter	WBTC	WETH
Benchmark drift μ_i (sample, incl. carry)	-0.314	-0.417
Log-price drift μ_i^X	-1.191	-1.356
Model log-return drift $\mathbb{E}[\mathrm{d}X_i]/\mathrm{d}t$	-0.355	-0.506
Realised (raw) log drift, price only	-0.390	-0.578
ECF drift normalizer (not used)	0.302	1.161
Diffusion volatility σ_i	0.264	0.328
Jump intensity λ_i	358.29	450.65
Upward-jump probability p_i	0.770	0.457
Mean upward / downward jump $\eta_{i,+}$ / $\eta_{i,-}$	0.00413 / 0.00367	0.01126 / 0.00601
Initial price $S_{i,0}$	64,125.20	1,872.81
Brownian correlation ρ; threshold b; horizon T	0.847; 0.78; $1/12$	

Table 2: Benchmark parameters: 840 four-hour WBTC/WETH log-returns, Aave v3 Ethereum, 2026-03-17 to 2026-08-04. Diffusion, jump and correlation parameters from the penalised ECF on EWM-de-meaned returns (span 30); drifts are the window's sample price-growth rates plus the average Aave rates (0.0061% WBTC supply, 2.5104% WETH borrow). Prices are oracle-implied USD values (reserve TVL / supplied balance); initial prices are the last sample (2026-08-04 16:00 UTC). Scripts `calibrate_btc_eth.py -merged` and `build_benchmark_params.py`.

value 0.60532: it is within about one standard error from 500 steps on (z between -1.1 and -0.2), and about two standard errors high at 125 and 250 steps, a residual discretisation effect. The naive check is biased by $+0.014$ at 125 steps and still $+0.004$ at 4,000. The analytic survival probability is therefore unbiased, and the bias of simulation is one-sided, as the missed intra-step crossings imply. Talbot inversion is also stable: varying M from 16 to 48 changes $p_{\mathrm{surv}}(h_0{=}0.07)$ by less than 10^{-8}. At 500 steps, 200,000 bridge-corrected paths take about 22 s for a standard error of 10^{-3}, against 23 ms for the noise-free Laplace survival probability: three orders of magnitude, and without sampling error.

6.2. Parameter sensitivity

Figure 1 sweeps each parameter in turn. At $H_0 = 1.2$ (benchmark survival 0.95) the *horizon* matters most: survival is 1.00 at one week and 0.80 at three months. *Jump intensity* and *diffusion volatility* come next: scaling the intensities from 0.1 to 2 times the benchmark moves survival from 1.00 to 0.88, and scaling σ_1, σ_2 from 0.3 to 2 times moves it from 0.97 to 0.87. Jumps carry about three quarters of the spread variance at the benchmark, which is why the intensity sweep matters as much as the volatility sweep. *Correlation* acts only through the spread variance: lowering ρ from 0.85 to 0.5 cuts survival to 0.89, and the conditional mean barely moves. Jump intensity and horizon also raise the conditional mean at low H_0 through survivor selection. Survival does not depend on b at fixed H_0 — only the payoff kernel does — so row (e) plots leverage $L_0 = H_0/(H_0 - b)$, which reaches $\approx 6.3\times$ at $H_0^{\min}$ for $b = 0.90$.

7. Liquidation Recovery, Overshoot and Log Wealth

This section supplies R_τ in (2) under the protocol's liquidation convention, at the cost of one additional Laplace inversion per moment order and with the barrier system unchanged. The same objects then give the law of the overshoot, the probability of bad debt and the expected log of terminal wealth.

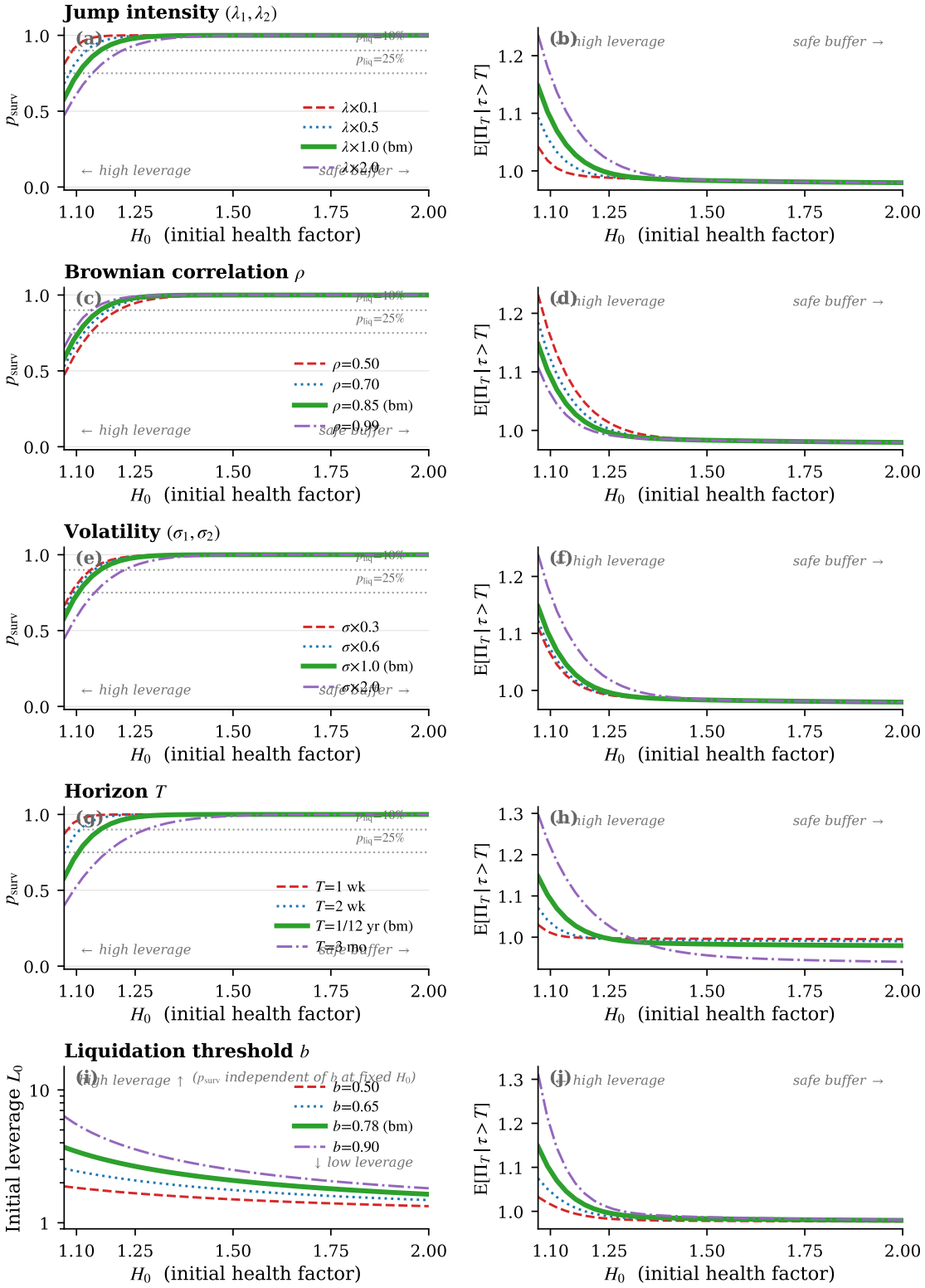


Figure 1: Left: survival probability (rows a–d) and initial leverage L_0 (row e, log scale) versus H_0. Right: conditional mean $\mathbb{E}[\Pi_T \mid \tau > T]$. Each row varies one parameter, others at Table 2; the heavier solid line is the benchmark. $H_0 \in [1.068, 2.000]$, 40 points per curve.

		p_{surv}					
h_0	H_0	Lap.	MC	Mean	Variance	Skewness	Ex. Kurt.
0.07	1.07	0.6053	0.6190	1.1386	0.0387	0.4976	0.1675
0.10	1.11	0.7545	0.7628	1.0840	0.0367	0.4508	0.0818
0.25	1.28	0.9906	0.9912	0.9914	0.0282	0.1146	−0.0215
0.50	1.65	1.0000	1.0000	0.9817	0.0149	0.1363	0.1565
0.75	2.12	1.0000	1.0000	0.9791	0.0103	0.2234	0.1448
1.00	2.72	1.0000	1.0000	0.9776	0.0085	0.2592	0.1445
1.50	4.48	1.0000	1.0000	0.9760	0.0072	0.2725	0.1421
2.00	7.39	1.0000	1.0000	0.9752	0.0068	0.2699	0.1373
∞	∞	1.0000	—	0.9742	0.0065	0.2627	0.1286

Table 3: Laplace–resolvent survival and survival-conditional statistics for the calibrated WBTC/WETH book. MC uses 10^5 Euler–Poisson paths with 252 steps over the month (seed 42). The bottom row is the analytical no-shorting limit; Laplace at $h_0 = 100$ matches it to relative error below 3×10^{-6}.

Steps	bridge-corrected p_{surv}	naive grid p_{surv}	naive bias
125	0.6077 (z=−2.14)	0.6198	+0.0145
500	0.6061 (z=−0.69)	0.6129	+0.0076
1,000	0.6061 (z=−0.71)	0.6122	+0.0068
2,000	0.6055 (z=−0.15)	0.6117	+0.0064
4,000	0.6065 (z=−1.12)	0.6092	+0.0039

Table 4: Survival at $h_0 = 0.07$ against steps over the month; 200,000 paths, MC standard error $\approx 1.1 \times 10^{-3}$, Laplace reference 0.60532; z is the standardised gap to Laplace. Script `mc_convergence.py`.

7.1. Recovery at the barrier

A liquidator repays debt and seizes collateral worth $(1+\ell)$ times the repayment, ℓ being the liquidation bonus (Aave v3 Ethereum: 0.05 for WETH, 0.045 for major stablecoins, 0.05–0.075 for wrapped BTC). Full repayment leaves $R_\tau = V_{1,\tau} - (1+\ell)V_{2,\tau}$. On a continuous crossing $V_{2,\tau} = bV_{1,\tau}$, so

$$R_\tau \;=\; V_{2,\tau}\,\frac{1-b(1+\ell)}{b}, \tag{32}$$

positive when $\ell < 1/b - 1$ (0.282 at $b = 0.78$). Writing recovery through the *debt* leg makes it proportional to $e^{X_{2,\tau}}$, the factor the framework already tilts by.

Remark 6 (Continuous-crossing approximation). A downward jump can carry H strictly below one, leaving less recovery than (32), possibly negative (bad debt). Equation (32) is therefore an upper bound; Section 7.3 computes the exact value.

7.2. The first-passage transform

We need $\mathbb{E}[e^{kX_{2,\tau}}\mathbf{1}_{\{\tau\le T\}}]$ for $k = 1, 2$. Since $\exp(kX_{2,t} - t\Psi_k)$, $\Psi_k := \Psi(0, -ik)$, is a martingale, changing measure on $\mathscr{F}_\tau$ gives $\mathbb{E}[e^{kX_{2,\tau}}\mathbf{1}_{\{\tau\le T\}}] = \mathbb{E}^{(k)}[e^{\Psi_k\tau}\mathbf{1}_{\{\tau\le T\}}]$, with Laplace transform

$$\int_0^\infty e^{-qT}\,\mathbb{E}\Big[e^{kX_{2,\tau}}\mathbf{1}_{\{\tau\le T\}}\Big]\,\mathrm{d}T = \frac{1}{q}\Big(1 - (q-\Psi_k)\,\widehat{p}^{(k)}_{\text{surv}}(q-\Psi_k)\Big), \qquad q > \Psi_k, \tag{33}$$

where $\widehat{p}^{(k)}_{\mathrm{surv}}$ is the survival transform of equation (31) under the k-tilted dynamics. Because Π_T and $R_\tau \mathbf{1}_{\{\tau\le T\}}$ live on disjoint events,

$$\mathbb{E}[W_T] = \mathbb{E}[\Pi_T] + \mathbb{E}[R_\tau \mathbf{1}_{\{\tau\le T\}}], \qquad \mathbb{E}[W_T^2] = \mathbb{E}[\Pi_T^2] + \mathbb{E}[R_\tau^2 \mathbf{1}_{\{\tau\le T\}}]. \tag{34}$$

At $k=0$, (33) returns p_{liq} and agrees with equation (31) to 1.7×10^{-11}.

Recovery is material. At $H_0 = 1.16$ ($p_{\mathrm{liq}} = 0.10$) it raises mean terminal wealth from 0.927 to 0.981 and cuts its variance from 0.127 to 0.053 by removing the total-loss atom at zero. It matters for every levered buffer, but not for the break-even of Section 8, which is set at zero debt, where liquidation vanishes.

7.3. Exact recovery under jump overshoot

With the actual leg values at liquidation and $h_\tau := h_0 + Z_\tau \le 0$ (zero at a continuous crossing),

$$R_\tau \;=\; V_{2,\tau}\left(\frac{e^{h_\tau}}{b} - (1+\ell)\right), \tag{35}$$

which is smaller than (32) whenever $h_\tau < 0$. Rather than integrating against the (exponential-mixture) overshoot density, we note that per unit initial equity

$$R_\tau = A\,e^{X_{1,\tau}} - B\,e^{X_{2,\tau}}, \qquad \mathrm{Hit}(a_1,a_2) := \mathbb{E}\left[e^{a_1X_{1,\tau}+a_2X_{2,\tau}}\mathbf{1}_{\{\tau\le T\}}\right], \tag{36}$$

with $A = H_0/(H_0-b)$ and $B = (1+\ell)b/(H_0-b)$, so $\mathbb{E}[R_\tau\mathbf{1}] = A\,\mathrm{Hit}(1,0) - B\,\mathrm{Hit}(0,1)$ and $\mathbb{E}[R_\tau^2\mathbf{1}] = A^2\mathrm{Hit}(2,0) - 2AB\,\mathrm{Hit}(1,1) + B^2\mathrm{Hit}(0,2)$. The tilted first-passage law already integrates over the landing point, so overshoot is captured exactly. Tilting by $\exp(a_1X_{1,t}+a_2X_{2,t}-t\Psi_{\mathrm{M}}(a_1,a_2))$, with $\Psi_{\mathrm{M}}(a_1,a_2) := \log\mathbb{E}[e^{a_1X_{1,1}+a_2X_{2,1}}]$, gives

$$\int_0^\infty e^{-qT}\,\mathrm{Hit}(a_1,a_2)\,\mathrm{d}T = \frac{1}{q}\Big(1-(q-\Psi_{\mathrm{M}})\,\widehat{p}^{(a_1,a_2)}_{\mathrm{surv}}(q-\Psi_{\mathrm{M}})\Big), \qquad q>\Psi_{\mathrm{M}}(a_1,a_2), \tag{37}$$

where $\widehat{p}^{(a_1,a_2)}_{\mathrm{surv}}$ is the survival transform under the (a_1,a_2)-tilted dynamics of Z, with exponent $\psi_Z^{(a_1,a_2)}(s) = \Psi_{\mathrm{M}}(a_1+s,a_2-s) - \Psi_{\mathrm{M}}(a_1,a_2)$ and phase rates $r_{1,+} = 1/\eta_{1,+} - a_1$, $r_{1,-} = 1/\eta_{1,-} + a_1$, $r_{2,+} = 1/\eta_{2,-} + a_2$, $r_{2,-} = 1/\eta_{2,+} - a_2$; $(a_1,a_2) = (0,k)$ recovers Lemma 3.1. The barrier system stays 3×3. $\mathrm{Hit}(0,k)$ reproduces (33) to 6×10^{-14} and $\mathrm{Hit}(0,0)$ reproduces $1-p_{\mathrm{surv}}$ to 3×10^{-11}; against an MC recording leg values at the crossing step, all five moments agree up to a discretisation bias that shrinks with the step count.

At the headline calibration's jump sizes ($\eta\approx0.4$–1.1%) the correction is small: at $H_0=1.16$ mean recovery falls from 0.054 to 0.052 (-3%) and wealth variance rises from 0.053 to 0.054. To see when overshoot bites, Table 5 scales the jumps two ways. Design A scales the mean jump size by s at fixed intensity, which multiplies jump variance by s^2 — a far more volatile market. The correction reaches -59% at $s=10$ and mean recovery turns negative at $s=20$: the continuous formula reports positive recovery where the protocol absorbs bad debt. Design B also divides intensity by s^2, holding jump variance fixed — genuinely rarer, larger jumps. The correction is just as severe (-49% at $s=10$, and at $s=20$ mean recovery is zero). The driver is jump size rather than jump variance: at fixed variance the correction still eliminates recovery, while the headline jumps carry about three quarters of the spread variance yet give only -3% because each is small. Throughout, R_τ is the signed residual equity; a non-recourse borrower receives R_τ^+, which differs from it only on bad-debt paths, whose probability Section 7.4 gives in closed form. The headline calibration produces its kurtosis from many small jumps (cf. the identification ridge of Ramezani and Zeng 2007 and Appendix B.2), so there the correction is small. It is not small under every calibration: the full-history daily fit of Appendix C, whose jumps are fewer and larger ($\eta_{2,+}\approx3.2\%$), gives a correction of -11% at $H_0=1.16$.

	Design A: variance $\propto s^2$		Design B: variance fixed	
Jump scale s	$\mathbb{E}[R_\tau]$ exact	rel. corr.	$\mathbb{E}[R_\tau]$ exact	rel. corr.
$1\times$ (fitted)	0.052	-3%	0.052	-3%
$3\times$	0.210	-16%	0.047	-11%
$10\times$	0.167	-59%	0.018	-49%
$20\times$	-0.089	-120%	-0.000	-103%

Table 5: Exact mean recovery at $H_0 = 1.16$ and its correction relative to the continuous-crossing value (32). The continuous-crossing *expected* recovery still moves with p_{liq}: under Design A p_{liq} rises from 0.10 to 0.69 (continuous mean recovery $0.054 \to 0.451$); under Design B it falls to 0.03 ($\to 0.019$). Script `overshoot_table.py`.

7.4. The overshoot law and the probability of bad debt

The joint hitting moments integrate over the landing point without exposing it. The landing point itself has a simple law. Write $u := -h_\tau \geq 0$ for the overshoot of the log health factor below the barrier.

Proposition 7.1 (Overshoot law)**.** *For every T there are weights $P_0, P_1, P_2 \geq 0$, with $P_0 + P_1 + P_2 = p_{\text{liq}}(T)$, such that for every bounded measurable f*

$$\mathbb{E}\big[f(u)\,\mathbf{1}_{\{\tau\leq T\}}\big] = P_0\,f(0) + \sum_{i=1}^{2} P_i \int_0^\infty f(u)\,r_i e^{-r_i u}\,\mathrm{d}u, \qquad r_1 = \frac{1}{\eta_{1,-}}, \quad r_2 = \frac{1}{\eta_{2,+}}. \tag{38}$$

The weights solve the 3×3 system $e^{jh_0}\mathrm{Hit}(j,-j) = P_0 + P_1\,r_1/(r_1+j) + P_2\,r_2/(r_2+j)$ for $j = 0,1,2$.

▷ *Proof, p. 35*

Z reaches the barrier either continuously (the diffusion creeps across, $u = 0$) or by a downward jump, which is a down-jump of X_1 (rate r_1) or an up-jump of X_2 (rate r_2). By memorylessness, the excess of a crossing jump over the distance to the barrier is exponential with the rate of that jump, whatever the path before it (Kou, 2002; Kou and Wang, 2003); Appendix E makes this precise with the compensation formula. This gives (38). Taking $f(u) = e^{-ju}$, with $e^{-ju} = e^{jh_0}e^{jZ_\tau}$ and Hit from (37), yields the system. At the benchmark the weights reproduce the hitting moments at $j = 5$ and $j = 10$ to about 10^{-10}, which checks the structure independently.

Recovery is non-positive exactly when $u \geq u^\star := -\log\big(b(1+\ell)\big)$ (0.200 at $b = 0.78$, $\ell = 0.05$). On those paths the protocol absorbs bad debt and a non-recourse borrower ends with nothing. Proposition 7.1 gives the bad-debt probability in closed form,

$$\mathbb{P}(R_\tau \leq 0,\ \tau \leq T) = P_1 e^{-r_1 u^\star} + P_2 e^{-r_2 u^\star}. \tag{39}$$

It is 1×10^{-9} at $H_0 = 1.16$ under the headline calibration, where jumps are below one percent, and 3×10^{-4} under the full-history daily calibration.

7.5. Expected log wealth

Moments of every admissible order cover polynomial criteria. The log of wealth, which defines the growth-optimal (Kelly) investor (Kelly, 1956; MacLean et al., 2011), is not polynomial, but it is also within reach. Per unit equity, $\Pi_T = e^{X_{2,T}}(e^{h_T} - b)/(H_0 - b)$ on survival and $R_\tau = e^{X_{2,\tau}}(e^{-u} - c)/(H_0 - b)$ on liquidation, with $c := b(1+\ell)$. Since $\log(e^h - b) = h - \sum_{n\geq 1}(b^n/n)e^{-nh}$ for $h \geq 0$,

$$\begin{aligned}\mathbb{E}[\log W_T] = \mathbb{E}\big[(X_{2,T} + h_T)\mathbf{1}_{\{\tau>T\}}\big] - \sum_{n\geq 1}\frac{b^n}{n}\,e^{-nh_0}S(-n,n) - \log(H_0 - b)\\ + \mathbb{E}\big[X_{2,\tau}\mathbf{1}_{\{\tau\leq T\}}\big] + \mathbb{E}\big[\log(e^{-u} - c)\,\mathbf{1}_{\{\tau\leq T\}}\big],\end{aligned} \tag{40}$$

where $S(a_1,a_2) := \mathbb{E}[e^{a_1 X_{1,T}+a_2 X_{2,T}}\mathbf{1}_{\{\tau>T\}}]$ is a killed joint moment. Its Laplace transform is the tilted survival transform at $q-\Psi_{\mathrm{M}}(a_1,a_2)$, by the change of measure behind (37) applied on $\mathscr{F}_T$. The linear terms are tilt derivatives of S and Hit at zero, and the last term is a one-dimensional integral against (38). The series converges geometrically because $be^{-h_T} \le b < 1$ on survival. The tilt $(-n,n)$ is admissible only for $n < \min(1/\eta_{2,+}, 1/\eta_{1,-})$, which caps the series under heavy jump tails. Since $\mathbb{E}[e^{-nh_T}\mathbf{1}_{\{\tau>T\}}]$ decreases in n, the truncation error after N terms is at most $\mathbb{E}[e^{-Nh_T}\mathbf{1}_{\{\tau>T\}}]\, b^{N+1}/((N+1)(1-b))$, which is below 10^{-5} in every case reported. The last term of (40) is taken over paths without bad debt; the next paragraph treats the rest.

Two features matter. First, under the zero-payoff convention $\log W_T = -\infty$ on every liquidated path, so a Kelly investor never levers; recovery is what makes the criterion usable. Second, on bad-debt paths $W_T = 0$ even with recovery, so strict Kelly is $-\infty$ at every levered buffer whenever (39) is positive. On those paths we replace $\log(e^{-u}-c)$ in (40) by $\log 10^{-4}$, which floors wealth at $10^{-4}e^{X_{2,\tau}}/(H_0-b)$, and report (39) alongside. Kelly quantities below refer to this floored objective. The floor is immaterial at the headline calibration and of order 10^{-3} in $\mathbb{E}[\log W_T]$ at the thinnest buffer under the full-history one. Evaluating (40) takes about 0.3 s per buffer. A bridge-corrected Monte Carlo that records the exact leg values at liquidation agrees to within 1.6 standard errors at $H_0 \in \{1.07, 1.16, 1.30\}$, and as $h_0 \to \infty$ the expression tends to $\mathbb{E}[X_{1,T}]$ to 10^{-6}.

8. Sizing on the Evaluation Map

The evaluation map is objective-independent. This section applies it to the sizing problem of Section 2.2: first under several decision rules at the benchmark drift, then by asking what the answer depends on, and finally by checking against realised outcomes. The drift is the input that matters, and it is kept outside the calibration throughout.

8.1. Decision rules on the map

Three rules illustrate how the same outputs enter different preferences. *A liquidation budget* selects the largest initial leverage whose liquidation probability does not exceed $\bar{p}$,

$$\max_{h_0 \in \mathscr{H}} L_0(h_0) \quad \text{subject to} \quad p_{\mathrm{liq}}(T,h_0) \le \bar{p}; \tag{41}$$

it needs only the survival probability and is the rule a risk desk would apply. *Mean–variance on terminal wealth*, (8) with exact recovery, and *Kelly*, $\max_{h_0}\mathbb{E}[\log W_T]$ (Section 7.5), trade expected wealth against risk. A fourth, popular rule — mean–variance on survival-conditional moments with a liquidation penalty — is not recommended: conditioning keeps only the paths that survived, so the conditional mean rises as the buffer thins and the rule rewards leverage for a survivorship artefact (Table 3).

The selections differ widely across rules (Table 6), as an objective-independent map should allow. Two features matter for what follows. First, the budget rule is robust to the drift, because over a month the liquidation probability is driven mainly by volatility and jumps: with a 5% budget it selects L_0 between 2.66 and 3.04 for any drift from -0.5 to $+0.6$ per year. The mean–variance and Kelly selections are driven by the drift: at the sample drift they lever moderately, and without a relative-drift view every one of them selects the unlevered position. Second, the search must include the unlevered position. A grid capped at a finite buffer, such as $H_0 \le 2$ ($L_0 = 1.64$), reports its cap whenever the true optimum is unlevered, which manufactures an apparent interior optimum. Fractional Kelly, the usual response to an uncertain drift (MacLean et al., 2011), scales the Kelly debt down but does not change whether to borrow at all: a fraction of zero debt is zero. The rest of this section therefore asks at which drift leverage starts to pay, and whether the data can tell.

Rule	Parameter	H_0	L_0	p_{liq}
Liquidation budget (41)	$\bar{p} = 0.01$	1.281	2.56	0.010
	$\bar{p} = 0.05$	1.199	2.86	0.050
	$\bar{p} = 0.10$	1.160	3.05	0.100
	$\bar{p} = 0.20$	1.118	3.31	0.200
Mean–variance on W_T	$\alpha = 0.5$	—	2.19	—
	$\alpha = 1$	—	1.91	—
	$\alpha = 2$	—	1.40	—
Kelly, $\mathbb{E}[\log W_T]$	full	—	1.91	—
	half	—	1.45	—
	quarter	—	1.23	—
Any of the last six, no relative-drift view	—	∞	1	0

Table 6: Selections on the benchmark map (sample drift $\nu = \mu_1 - \mu_2 = +0.103$ per year, $T = 1/12$). The budget rows maximise L_0 subject to $p_{\text{liq}} \le \bar{p}$. The mean–variance and Kelly rows search 30 debt levels over the full buffer range plus the unlevered position, with exact recovery; fractional Kelly holds the stated fraction of the Kelly debt. The last row sets the relative price drift to zero, so that ν equals the net carry. Scripts `decision_rules.py` and `breakeven_drift.py`.

8.2. Break-even drift views

Write $x := n_2 S_{2,0} = b/(e^{h_0} - b)$ for the debt per unit of equity, so that $x = 0$ is the unlevered position and $L_0 = 1 + x$. Leverage improves on the unlevered position exactly when the first unit of debt does, and that is a closed-form question. As $x \to 0$ the buffer becomes infinite and the liquidation probability vanishes, so $W_T = (1+x)e^{X_{1,T}} - x e^{X_{2,T}}$ to first order, and

$$\left.\frac{\mathrm{d}\Phi}{\mathrm{d}x}\right|_{x=0} = \left(\mathbb{E}e^{X_{1,T}} - \mathbb{E}e^{X_{2,T}}\right) - \alpha \operatorname{Cov}\left(e^{X_{1,T}},\, e^{X_{1,T}} - e^{X_{2,T}}\right), \qquad \left.\frac{\mathrm{d}\,\mathbb{E}[\log W_T]}{\mathrm{d}x}\right|_{x=0} = 1 - \mathbb{E}\, e^{X_{2,T} - X_{1,T}}, \quad (42)$$

for mean–variance (8) and for Kelly. All terms are joint exponential moments $e^{T\Psi_{\mathrm{M}}(a_1,a_2)}$. The *break-even drift view* $\nu^\star$ is the spread drift $\nu = \mu_1 - \mu_2$ (price growth plus net carry) at which the derivative vanishes, holding the level μ_2 at its sample value. Without liquidation both objectives are concave in x (mean–variance is a concave quadratic, log a concave function of an affine one), so the sign of the derivative at $x = 0$ decides between the unlevered position and some leverage. Liquidation breaks this argument, so we check it numerically: a search over 30 debt levels spanning the full buffer range, from $x = 10^{-3}$ ($H_0 \approx 781$) to the origination maximum, with exact recovery and the floored Kelly objective of Section 7.5, confirms the switch at $\nu^\star$ for every objective and calibration.

For Kelly the condition $\mathbb{E}e^{X_{2,T} - X_{1,T}} = 1$ is $\Psi_{\mathrm{M}}(-1, 1) = 0$, which does not depend on T and, with independent jump processes, solves in closed form:

$$\nu^\star_{\text{Kelly}} = \underbrace{\sigma_1^2 - \rho\sigma_1\sigma_2}_{\text{diffusion}} + \underbrace{\lambda_1\left[M_1(1) + M_1(-1) - 2\right]}_{\text{WBTC jumps}}, \qquad M_1(s) = \frac{p_1}{1 - s\eta_{1,+}} + \frac{1 - p_1}{1 + s\eta_{1,-}}. \quad (43)$$

The break-even views are small (Table 7): about 0.4–1.6% per year under the headline calibration and 2.4–11% under the full-history ones, close to linear in α, with Kelly near $\alpha = 1$. Equation (43) says why. The diffusion part is negative, because WBTC is the less volatile leg ($\sigma_1 < \rho\sigma_2$): at the margin, shorting WETH against WBTC reduces diffusive risk. The break-even is positive only through WBTC's own jumps, which the short leg does not hedge: $-0.004 + 0.012$ under the headline calibration, $-0.017 + 0.065$ under the full-history daily one and $-0.009 + 0.054$ under the 2024 one. The second term rests

on the jump parameters, which the data identify weakly (Appendix B), and a common jump component, which the model excludes, would reduce it by the covariance of the two legs' jumps. Section 8.4 shows that this is the case in the data: the realised break-even of WBTC/WETH is negative. With no relative-drift view at all, ν equals the net carry ($\approx -2.5\%$), below every break-even, so the unlevered position is optimal for every objective. Above $\nu^\star$ the optimal leverage rises quickly with the view (Figure 2).

Calibration	model $\bar{\sigma}_Z$	$\alpha = 0.5$	$\alpha = 1$	Kelly	$\alpha = 2$
Headline, 4h, 2026-03 to 08 (0.38 y)	0.335	+0.004	+0.008	+0.008	+0.016
Full history, daily (3.7 y)	0.478	+0.027	+0.054	+0.048	+0.110
2024 onward, 4h (2.8 y)	0.485	+0.024	+0.049	+0.045	+0.099

Table 7: Break-even spread-drift views $\nu^\star$ (per year, price growth plus net carry): the unlevered position is optimal below $\nu^\star$ and some leverage above it, for mean–variance on terminal wealth with exact recovery and for Kelly, from (42), with μ_2 at the sample level of the matching window (the full-history one for the 2024 calibration). "Model $\bar{\sigma}_Z$" is the calibrated total (diffusion plus jump) spread volatility. Script `breakeven_drift.py`.

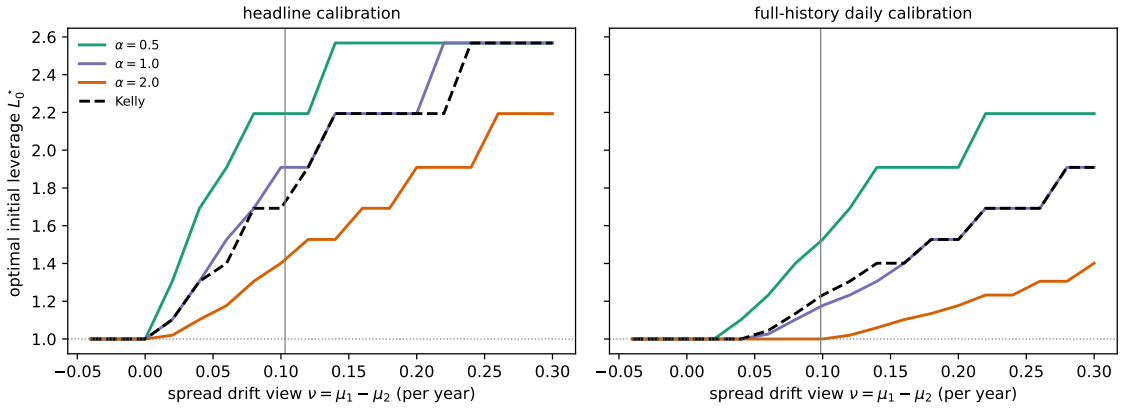


Figure 2: Optimal initial leverage $L_0^\star$ against a drift view $\nu = \mu_1 - \mu_2$, mean–variance on terminal wealth ($\alpha = 0.5, 1, 2$) and Kelly, exact recovery, over the full buffer range. Left: headline calibration; right: full-history daily calibration. The optimum is unlevered ($L_0^\star = 1$) below the break-even view and rises quickly above it. The vertical line marks the sample drift of each window. The optimum is located on 30 debt levels plus the unlevered position, hence the steps. Script `breakeven_drift.py`.

8.3. What the data say about the drift

The spread drift has two parts, $\nu = \nu_{\text{price}} + c$. The net carry $c = r_s - r_b$ is observed on-chain, about -2.5% per year for WBTC/WETH. The relative price growth ν_{price} is the expected outperformance of WBTC over WETH, and it is estimated poorly at any sampling frequency, because the standard error of a drift depends only on calendar length (Merton, 1980), so, as in Section 6, we use sample means. The headline window gives $\nu = +0.10$ (price $+0.13$, carry -0.025) with a Newey–West standard error of 0.34; the full 3.7-year daily history gives $\nu = +0.10$ (price $+0.13$, carry -0.03) with a Newey–West and block-bootstrap standard error of 0.19; and the yearly estimates of ν_{price} range from -0.16 to $+0.34$ with standard errors of 0.3–0.5. Both point estimates lie above every headline break-even and above most full-history ones, and at them the optimal leverage is moderate (1.4–2.2 under the headline calibration,

1.0–1.5 under the full-history ones). But the standard errors are several times the gap between estimate and break-even: telling $+0.10$ from the full-history Kelly break-even of $+0.05$ at two standard errors would need a standard error of 0.025, about two centuries of data. Both reserves accept collateral and borrowing, so the book can also be held the other way round. Reversing the legs flips the sign of ν_{price} and changes the carry: over the full history, WETH collateral against WBTC debt has $\nu_{\text{price}} = -0.13$ and $c = +1.1\%$, so $\nu = -0.11$. Its Kelly break-even, the mirror image of (43) with the legs exchanged, is $+0.10$ to $+0.19$ across the calibrations and has a positive diffusion part, because WETH is the more volatile leg; the reversed book is thus far from paying, even though WETH's higher liquidation threshold (0.83 against 0.78) would allow more leverage. A relative-drift view therefore chooses the orientation as well as the buffer.

A carry-anchored Bayesian layer turns this into a probability. Take $\nu \sim \mathcal{N}(c, 0.20^2)$, the view that relative prices have no predictable drift, and update it with the sample estimate. The posterior probability that leverage pays, $\mathbb{P}(\nu > \nu^\star)$, is 48–51% for every objective under the headline calibration and 30–54% under the full-history ones, treating $\nu^\star$ as known. Under a flat prior, which takes the sample estimate at face value, it is 60–62% and 48–65%. Either way the data leave the question close to open. Maximising the posterior expectation of $\mathbb{E}[\log W_T]$ (Bayes–Kelly) has an exact rule at the margin: with posterior $\mathcal{N}(m, s^2)$, lever if and only if $m > \nu^\star_{\text{Kelly}} + \frac{1}{2} T s^2$. The variance term is small (0.001 at the headline posterior), so the rule is decided by the posterior mean. The carry-anchored prior pulls it to $+0.008$ (headline) and $+0.039$ (full history), just below the thresholds $+0.009$ and $+0.046$–0.049, and Bayes–Kelly selects the unlevered position under all three calibrations. Under the flat prior $m = \hat{\nu} \approx 0.10$ and it levers. The unlevered choice therefore reflects the prior's shrinkage toward zero relative drift, not estimation risk as such. Out of sample, the fitted ECF drift also worsens the map's liquidation forecasts relative to setting $\nu_{\text{price}} = 0$ (Appendix C).

8.4. Did leverage pay? A realised backtest

The break-even rule says when leverage pays; whether it did is a question for realised data, and one that needs no model. We replay every pair of the pre-registered cross-section (Appendix D.2; 37 pairs) hour by hour on the Aave v3 Ethereum history: oracle-implied prices, the collateral's supply rate and the debt's borrow rate. Windows run from the first hour at which both reserves are live (27 January 2023 for the original reserves, up to 23 September 2024 for cbBTC) to 13 October 2026. Equity is in dollars. At each roll date the book is reset to initial leverage L_0; between rolls interest accrues on both legs, and whenever the health factor falls below one at an hourly mark a keeper repays half the debt (all of it below 0.95) and seizes collateral worth the repayment plus the reserve's bonus, as on Aave v3. The benchmark holds the collateral unlevered. Rolled books are averaged over four roll phases and buy-and-hold books over four start dates 120 days apart. Standard errors are Newey–West on daily excess log returns (60-day lag).

Roll horizon	$L_0 = 1.25$	$L_0 = 1.5$	$L_0 = 2$	$L_0 = 3$
1 day	-0.038 (4)	-0.102 (4)	-0.287 (3)	-0.768 (0)
30 days	-0.042 (4)	-0.113 (4)	-0.331 (4)	-0.994 (1)
365 days	-0.008 (5)	-0.112 (5)	-0.559 (3)	-0.889 (2)
Buy and hold	-0.017 (12)	-0.036 (12)	-0.303 (10)	-0.451 (7)

Table 8: Realised excess log growth per year of the levered over the unlevered book: median over the 37 pairs, with the number of pairs in which leverage paid in parentheses. Aave v3 liquidation rules, current reserve parameters, no trading costs. Script `realised_backtest.py`.

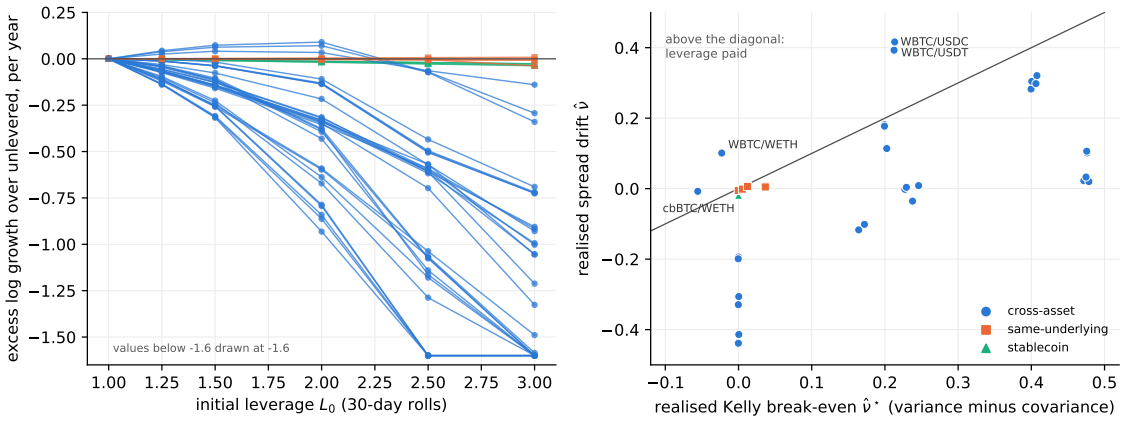


Figure 3: Left: realised excess log growth against initial leverage, 30-day rolls, one line per pair. Right: realised spread drift $\hat{\nu}$ against the realised Kelly break-even $\hat{\nu}^\star$, both from daily gross returns including interest; above the diagonal a little leverage paid. Script `fig_realised_backtest.py`.

Leverage rarely paid (Table 8, Figure 3). At $L_0 = 1.25$ it beat the unlevered book in 4 of 37 pairs — WBTC against USDC, USDT and WETH, and cbBTC against WETH, all books long BTC in a window in which BTC rose 3.6-fold and doubled against ETH — and in none significantly. Of 1,332 pair, leverage and horizon combinations, one has $t > 2$ and 338 have $t < -2$; the four stablecoin pairs lose significantly at every leverage up to $3\times$ and every horizon, because their net carry over the window is negative and nearly deterministic. Losses grow quickly with leverage, through volatility drag at every horizon and through liquidations once the book is left for a week or more: from $L_0 = 2.5$ most pairs are liquidated at least once at those horizons, and a book rolled at the maximum LTV loses in every pair. A 10 bp trading cost per reset leaves these numbers almost unchanged, and so does the paper's full-liquidation convention for rolled books up to $L_0 = 2$; for buy-and-hold at high leverage it loses less than Aave's partial liquidations, because after one liquidation the residual is held unlevered. Rolling more often does not help: at low leverage the excess is about the same from one day to a year. Buy and hold does better, because a static book's leverage falls as it gains and it avoids the volatility drag of re-levering, but it still loses in most pairs. Nor is short-term profitability a matter of calendar time: at $L_0 = 2$ and 30-day rolls, 17% of cross-asset pair-years were positive, almost all of them BTC-long books in 2023–24 or ETH-long books against dollars in 2023, and none of the 29 in 2025. The window is one cycle — a bull market to the 2025 peaks, then a decline of 53% for BTC and 68% for ETH to mid-2026 — with one dominant cross-asset trend, so these numbers describe this history, not a law. A simple ex-ante rule that levers to $2\times$ for a month only after a positive trailing-year spread return lost less than always levering in 29 pairs but beat the unlevered book in only 5.

The realised break-even confirms the rule and exposes the model. For a daily-rolled book the marginal log growth of the first unit of debt is $\mathbb{E}[1 - R_2/R_1]$ in daily gross returns, so leverage pays at the margin if and only if $\hat{\nu} > \hat{\nu}^\star$, the realised counterpart of (43); this is a first-order identity, and it sorts all 37 pairs correctly. Its value lies in the comparison with the model. For WBTC/WETH, $\hat{\nu}^\star = -0.023$ (daily; -0.026 four-hourly), against the full-history model's $+0.048$. The legs' realised correlation is 0.81, and on 85% of the days with a three-standard-deviation move in either leg both legs move by more than one and a half standard deviations in the same direction; the calibrated model, whose jumps are independent, has correlation 0.50. With only Brownian correlation to carry co-movement, the fit cannot match the common jumps: it understates both legs' variances (WBTC 0.15 against 0.21 per year) and, by much more, their covariance (0.11 against 0.24), so the spread variance is overstated (0.23 against 0.14) and

the break-even, collateral variance minus covariance, turns positive. The same overstatement inflates the model's liquidation probabilities and is consistent with their excess over realised frequencies out of sample (Appendix C). A common jump component — a third Poisson process moving both legs, with double-exponential marks so that the spread's jumps stay in the mixed-exponential class of Section 5, at the cost of more phases and a larger barrier system — is therefore the most consequential extension of the model.

8.5. A decision-theoretic extension

These results suggest a separate treatment of the sizing problem, which we leave to future work. Its question is whether a leveraged position of this kind can be sized from data at all. Four pieces would be needed. *(i)* A preference-free break-even: the killed law of Z_T and the overshoot law of Proposition 7.1 determine the full distribution of W_T, so one can find the drift views below which the unlevered position second-order stochastically dominates every levered one. Below that view every risk-averse investor agrees, not only the objectives above. *(ii)* The CRRA family, which extends Section 7.5 by the binomial series in place of the logarithmic one, with the same tilted moments. *(iii)* An explicit account of estimation risk in the tradition of Barberis (2000), Kan and Zhou (2007) and DeMiguel et al. (2007), and of fractional Kelly (MacLean et al., 2011), with the break-even view $\nu^\star$ as the decision rule: lever only if one's relative-drift view, including carry, exceeds it, and size the position by the posterior rather than the point estimate. *(iv)* A repeated, monthly-rolled version in which liquidation is partial ruin and long-run growth is the natural criterion.

9. Conclusion

We derive a semi-analytic evaluation map for leveraged DeFi long–short positions under liquidation and jump risk. The initial log health factor reduces the holdings to one parameter, an exponential change of measure reduces each admissible payoff moment to a killed expectation of the log price ratio, and under Kou dynamics three barrier conditions give a 3×3 resolvent system for every moment order. Bivariate tilts of the same system give the exact liquidation recovery under jump overshoot, the overshoot law and the probability of bad debt, and the expected log of terminal wealth. The map is noise-free, fast and validated against an accuracy-matched simulator; out of sample its one-month liquidation probabilities lie above realised frequencies, systematically so for some predictors.

Applied to sizing, the map turns any of the usual objectives into a one-dimensional search, and the selection moves with the objective, its risk parameters and the loss convention. Beneath that variety sits one quantity: the closed-form break-even drift above which the first unit of debt pays, about 0.4–11% per year depending on risk attitude and calibration. The sample drift lies near or above it, but with standard errors several times the gap, so the data cannot tell an investor whether to lever. Ex post, leverage rarely paid: replayed under Aave's liquidation rules over 2023–2026, it beat the unlevered position in 4 of 37 pairs at modest leverage and in none significantly. The realised break-evens also disagree with the model's for WBTC/WETH, because the model's independent jumps cannot reproduce the legs' common jumps.

The principal output is therefore not a leverage number but the map itself, which evaluates a buffer under a stated drift view or stress scenario and is the input any sizing rule needs. The most consequential extension is a common jump component. Others are a decision-theoretic treatment of sizing under drift uncertainty (Section 8.5), an endogenous utilization process in place of the exogenous rate regime of Appendix D, the stochastic control problem of Remark 2, and a market-impact channel that makes the liquidation bonus endogenous in the overshoot regime.

Appendix A: Killing at the Barrier and Downward-Phase Corrections

Section 5 uses only the generator $\mathscr{L}^{(k)}$ from Lemma 3.1, with a killed expectation extended by zero below the barrier. This appendix records how that convention changes the downward jump integrals and derives the three rows of the barrier system (26).

A.1. Phase Operators and Zero Extension

Definition A.1 (Phase operators)**.** For $j = 1, 2$, the phase operators associated with the tilted jump kernels of Z under $\mathbb{P}^{(k)}$ are

$$K^{(k)}_{j,+}[f](x) := \int_0^\infty f(x+y) r^{(k)}_{j,+} e^{-r^{(k)}_{j,+} y}\, \mathrm{d}y, \qquad K^{(k)}_{j,-}[f](x) := \int_0^\infty f(x-y) r^{(k)}_{j,-} e^{-r^{(k)}_{j,-} y}\, \mathrm{d}y.$$

For any complex γ with $\mathrm{Re}(\gamma) < r^{(k)}_{j,+}$ and $\mathrm{Re}(\gamma) > -r^{(k)}_{j,-}$, direct integration gives

$$K^{(k)}_{j,+}[e^{\gamma\cdot}](x) = \frac{r^{(k)}_{j,+}}{r^{(k)}_{j,+} - \gamma} e^{\gamma x}, \qquad K^{(k)}_{j,-}[e^{\gamma\cdot}](x) = \frac{r^{(k)}_{j,-}}{r^{(k)}_{j,-} + \gamma} e^{\gamma x}. \tag{44}$$

This phase construction originates in Kou and Wang (2003) for first-passage problems under the double-exponential model and is used by Sepp (2004) to reduce barrier pricing to a finite linear system.

The tilted phase intensities are

$$\lambda^{(k)}_{1,+} = \lambda_1 p_1, \quad \lambda^{(k)}_{1,-} = \lambda_1 (1-p_1), \quad \lambda^{(k)}_{2,+} = \frac{\lambda_2 (1-p_2)}{1 + k\eta_{2,-}}, \quad \lambda^{(k)}_{2,-} = \frac{\lambda_2 p_2}{1 - k\eta_{2,+}}. \tag{45}$$

Using Definition A.1, the full-line action of the generator is

$$\mathscr{L}^{(k)} f(x) = \mu^{(k)}_Z f'(x) + \frac{\sigma_Z^2}{2} f''(x) + \sum_{a=1}^{2} \lambda^{(k)}_{a,+} \big(K^{(k)}_{a,+}[f](x) - f(x)\big) + \sum_{d=1}^{2} \lambda^{(k)}_{d,-} \big(K^{(k)}_{d,-}[f](x) - f(x)\big). \tag{46}$$

Applying (44) recovers the exponential symbol (17).

Now let f be defined on the alive half-line $x > 0$ and let $\bar{f} = f\mathbf{1}_{(0,\infty)}$ be its zero extension. Upward jumps remain in the alive region, whereas a downward jump of size $y \geq x$ lands in the absorbing region. Consequently,

$$K^{(k)}_{d,-}[\bar{f}](x) = K^{(k),D}_{d,-}[f](x) := \int_0^x f(x-y) r^{(k)}_{d,-} e^{-r^{(k)}_{d,-} y}\, \mathrm{d}y, \qquad x > 0. \tag{47}$$

Substitution into (46) gives

$$\begin{aligned} \mathscr{L}^{(k)}[\bar{f}](x) = \mu^{(k)}_Z f'(x) + \frac{\sigma_Z^2}{2} f''(x) + \sum_{a=1}^{2} \lambda^{(k)}_{a,+} \big(K^{(k)}_{a,+}[f](x) - f(x)\big) \\ + \sum_{d=1}^{2} \lambda^{(k)}_{d,-} \big(K^{(k),D}_{d,-}[f](x) - f(x)\big), \qquad x > 0. \end{aligned} \tag{48}$$

Thus killing changes the arguments on which $\mathscr{L}^{(k)}$ acts, not the identity of the generator. Equation (48) is the standard zero-extension construction for a Lévy process killed at an absorbing boundary.

A.2. Downward-Phase Residual and Barrier Conditions

Lemma A.1 (Truncated downward phase)**.** *For $r > 0$, $\beta \in \mathbb{C} \setminus \{-r\}$, and $x > 0$,*

$$\int_0^x e^{\beta(x-y)} r e^{-ry}\,\mathrm{d}y = \frac{r}{r+\beta} e^{\beta x} - \frac{r}{r+\beta} e^{-rx}. \tag{49}$$

No restriction $\mathrm{Re}(\beta) > -r$ is needed, since the integral is over a finite interval.

▷ *Proof, p. 34*

For a finite exponential sum $v(x) = \sum_{\beta \in \mathscr{E}} a_\beta e^{\beta x}$, define the truncation-correction coefficient

$$\mathfrak{C}_r[v] := \sum_{\beta \in \mathscr{E}} a_\beta \frac{r}{r+\beta}. \tag{50}$$

Then (49) gives

$$K^D_{r,-}[v](x) := \int_0^x v(x-y) r e^{-ry}\,\mathrm{d}y = \sum_{\beta \in \mathscr{E}} a_\beta \frac{r}{r+\beta} e^{\beta x} - \mathfrak{C}_r[v] e^{-rx}. \tag{51}$$

Combining (48) and (51), for exponents at which the upward integrals are finite and which avoid the downward phase poles,

$$\mathscr{L}^{(k)}[\bar{v}](x) = \sum_{\beta \in \mathscr{E}} a_\beta \psi_Z^{(k)}(\beta) e^{\beta x} - \sum_{d=1}^{2} \lambda_{d,-}^{(k)} \mathfrak{C}_{r_{d,-}^{(k)}}[v] e^{-r_{d,-}^{(k)} x}, \qquad x > 0, \tag{52}$$

where $\psi_Z^{(k)}$ is understood through the meromorphic continuation of its exponential symbol outside the common full-line moment strip.

Apply (52) term by term to the interior ansatz (24). Its forcing coefficients and characteristic-root equations give

$$(q - \mathscr{L}^{(k)})\widehat{U}_k(q,x) = G_k(x) + \sum_{d=1}^{2} \lambda_{d,-}^{(k)} \mathfrak{C}_{r_{d,-}^{(k)}}[\widehat{U}_k(q,\cdot)] e^{-r_{d,-}^{(k)} x}. \tag{53}$$

When the two downward rates are distinct, the residual exponentials are linearly independent, so the interior equation holds if and only if both correction coefficients vanish. Since $\sigma_Z > 0$, the lower boundary is regular for the Brownian component and absorption also imposes the Dirichlet condition. The three conditions are therefore

$$\widehat{U}_k(q,0) = 0, \qquad \mathfrak{C}_{r_{1,-}^{(k)}}[\widehat{U}_k(q,\cdot)] = 0, \qquad \mathfrak{C}_{r_{2,-}^{(k)}}[\widehat{U}_k(q,\cdot)] = 0. \tag{54}$$

Substituting (24) into (54) gives

$$\sum_{m=1}^{3} B_m^{(k)} = -\sum_{j=0}^{k} \frac{d_{k,j}}{q - \psi_Z^{(k)}(j)},$$

and, for $d = 1, 2$,

$$\sum_{m=1}^{3} \frac{r_{d,-}^{(k)}}{r_{d,-}^{(k)} + \gamma_{m+3}^{(k)}(q)} B_m^{(k)} = -\sum_{j=0}^{k} \frac{d_{k,j}}{q - \psi_Z^{(k)}(j)} \frac{r_{d,-}^{(k)}}{r_{d,-}^{(k)} + j}.$$

These are precisely the three rows of (26).

Remark 7 (Coincident downward rates). If $r_{1,-}^{(k)} = r_{2,-}^{(k)} =: r$, the two downward kernels combine into a single phase. The meromorphic symbol must first be reduced by cancelling the repeated phase factor; otherwise clearing denominators introduces a spurious root at the pole $-r$. The reduced characteristic equation has two genuine decaying roots, say $\gamma_4^{(k)}(q), \gamma_5^{(k)}(q)$, and the conditions $\widehat{U}_k(q,0) = 0$ and $\mathfrak{C}_r[\widehat{U}_k(q,\cdot)] = 0$ give the 2×2 system

$$\begin{pmatrix} 1 & 1 \\ \dfrac{r}{r+\gamma_4^{(k)}(q)} & \dfrac{r}{r+\gamma_5^{(k)}(q)} \end{pmatrix} \begin{pmatrix} B_1^{(k)} \\ B_2^{(k)} \end{pmatrix} = - \begin{pmatrix} \displaystyle\sum_{j=0}^{k} \frac{d_{k,j}}{q-\psi_Z^{(k)}(j)} \\ \displaystyle\sum_{j=0}^{k} \frac{d_{k,j}}{q-\psi_Z^{(k)}(j)} \dfrac{r}{r+j} \end{pmatrix}.$$

Thus the 3×3 description applies to the generic distinct-rate regime; the coincident-rate case is a genuine lower-dimensional phase model, not a singular generic system.

Appendix B: Empirical Calibration

B.1. Estimator

The model characteristic function over an observation interval Δ is available in closed form,

$$\phi_\theta(u,v;\Delta) = \exp\big(\Delta\,\Psi_\theta(u,v)\big). \tag{55}$$

Given N bivariate log-returns $(r_{1,n}, r_{2,n})$, with empirical characteristic function

$$\widehat{\phi}_N(u,v) = \frac{1}{N}\sum_{n=1}^{N} \exp\big(i(u\,r_{1,n} + v\,r_{2,n})\big), \tag{56}$$

we estimate θ by penalised ECF matching,

$$\begin{aligned}\widehat{\theta} = \arg\min_{\theta\in\Theta} &\sum_{(u,v)\in\mathscr{G}} w(u,v)\,\big|\widehat{\phi}_N(u,v) - \phi_\theta(u,v;\Delta)\big|^2 \\ &+ \kappa_\lambda \sum_{i=1}^{2}\big(\log\lambda_i - \log\bar{\lambda}_i\big)^2 + \kappa_\eta \sum_{i=1}^{2}\sum_{\pm}\big(\log\eta_{i,\pm} - \log\bar{\eta}_{i,\pm}\big)^2.\end{aligned} \tag{57}$$

The first term is the ECF distance of Singleton (2001), which can attain ML efficiency over a rich grid (Feuerverger and McDunnough, 1981) and avoids a two-dimensional density inversion inside every optimizer call (for univariate DEJD likelihood estimation see Ramezani and Zeng 2007). The two soft penalties, with $\kappa_\lambda = 0.05$ and $\kappa_\eta = 0.02$, anchor the intensities to a two-moment (variance and fourth-cumulant) estimate $\bar{\lambda}_i$ and the jump sizes to peak-over-threshold mean excesses $\bar{\eta}_{i,\pm}$. They exist to stop the optimiser sliding along the λ–η ridge discussed in Appendix B.2 toward very many, very small jumps. The fitted jump parameters are therefore partly anchored rather than purely ECF-determined, which should be kept in mind when reading them.

Grid. Frequencies are standardised: a dimensionless base value a maps to the raw frequency a/s, where s is the robust (IQR) scale of the relevant return series, so the same grid serves any sampling interval. Six groups are used: marginal directions for each asset (eleven base values from 0.1 to 10), same-sign and opposite-sign joint directions on a 6×6 grid ($a,b \in \{0.25, 0.5, 1, 2, 3, 5\}$) that identify ρ, the spread direction $(s,-s)$ scaled by the robust scale of Z ($a \in \{0.1,\ldots,5\}$), which matches the characteristic function of the liquidation-driving process directly, and a jump-tail group at high base frequencies ($a \in \{8, 12, 16, 20, 25\}$) on the marginal and spread directions. Each point enters with its conjugate, giving 234 points for 13 parameters. Weights are $w = \exp(-0.02(a^2+b^2))$ in the base frequencies, doubled on the spread group.

Optimisation. Constraints are enforced by reparameterisation ($\sigma_i, \lambda_i, \eta_{i,-}$ via exp; $p_i, \eta_{1,+}$ via sigmoid; ρ via tanh), with $\eta_{2,+} = \eta_{2,+}^{\max}\mathrm{sigmoid}(d_2)$, $\eta_{2,+}^{\max} = (1-\varepsilon)/K$, so that $K\eta_{2,+} < 1$ holds by construction; the headline bounds are $\lambda_i \le 5000$, $p_i \in [0.001, 0.999]$ and $\eta \ge 10^{-5}$. The non-convex objective is warm-started by a threshold decomposition in the spirit of Mancini (2009): returns far from the median in robust-scale units are flagged as jumps, σ_i and ρ come from the remaining observations, and μ_i from the sample drift with a damped mean-jump correction. L-BFGS-B is then run from a multi-start cloud (30 starts for the headline fit): the warm start, a set of ridge-traversing starts that scale λ by c and η by $1/\sqrt{c}$, and randomly perturbed scenario starts. The lowest objective is kept. The implementation is the `optimal_long_short.calibration` package; `jobs/calibrate_kou.py` demonstrates parameter recovery on synthetic data.

B.2. Identifiability and model comparison

Jump-diffusions have approximate ridges along directions preserving total jump variance $\lambda_i(p_i\eta_{i,+}^2 + (1-p_i)\eta_{i,-}^2)$ (Ramezani and Zeng, 2007). A circular block bootstrap (block length 12, $B = 200$, each resample recalibrated with the headline bounds and penalties from 8 starts; Table 9) shows the consequences. Diffusion volatilities and correlation are sharp (relative SEs of about 6% for σ_i and 2% for ρ). The drifts and the jump parameters are not: $\hat{\mu}_1 = 0.30 \pm 0.72$, $\hat{\mu}_2 = 1.16 \pm 1.17$, and their difference -0.86 has SE 1.06. These are the ECF drift normalizers, not sample drifts (Section 6), so their standard errors measure how loosely the ECF pins the drift parameters rather than the precision of a drift estimate. The intensities carry relative SEs near 30%, even with the soft anchors of (57) holding them toward the moment estimates, and the jump-size intervals span a factor of five to six for $\eta_{i,+}$ and about fifteen for $\eta_{i,-}$. The block length of two days is short relative to the mean reversion visible at a week in this window, so these standard errors may understate the uncertainty. No resample reaches the $\lambda \leq 5000$ bound. The data pin down σ_Z, not the drift or the jump decomposition.

Parameter	point	bootstrap SE	95% CI
σ_1	0.264	0.016	[0.228, 0.295]
σ_2	0.328	0.019	[0.296, 0.370]
ρ	0.847	0.020	[0.800, 0.878]
μ_1	0.302	0.725	[−0.61, 1.75]
μ_2	1.161	1.172	[−0.91, 3.58]
$\mu_1 - \mu_2$	−0.858	1.058	[−2.65, 1.36]
λ_1	358	106	[219, 628]
λ_2	451	134	[253, 770]
p_1	0.770	0.210	[0.24, 1.00]
$\eta_{1,+}$	0.0041	0.0036	[0.0015, 0.0085]
$\eta_{2,+}$	0.0113	0.0041	[0.0031, 0.0191]

Table 9: Block-bootstrap standard errors ($B = 200$), each resample recalibrated with the headline bounds ($\lambda \leq 5000$) and anchoring penalties. The two drift estimates are correlated (+0.46). Script `jobs/bootstrap_se.py`.

In the headline window the fitted jumps are tail-fattening, not rare and large. Fitting GBM, Merton (Gaussian jumps) and Kou to the ECF of the spread Z on a common grid, Kou has the smallest in-sample objective (1.7×10^{-6} against 1.3×10^{-5} and 3.4×10^{-4}), is preferred under a BIC-style penalty, and matches the empirical 1%/99% one-step quantiles (−1.59%/+1.43% against −1.52%/+1.42%). But it reproduces the excess kurtosis of about 3 with *many small* jumps — about 2,700 per year of mean size 0.2–0.4% in this univariate fit, and 350–450 per year of 0.4–1.1% in the bivariate headline fit — not with rare large moves; flash crashes enter only through the scaled experiments of Section 7.3 and the full-history calibration below. Out of sample (fit on the first 60%, score on the last 40%), Kou's edge over Merton all but vanishes (1.13 against 1.14×10^{-3}; GBM 1.60×10^{-3}).

Appendix C: Full History, Sampling Frequency and Out-of-Sample Validation

Data. The hourly snapshots extend back to the launch of Aave v3 on Ethereum, giving 3.7 years (2023-01-27 to 2026-10-13). A snapshot exists only in hours with market activity, so the thin early WBTC market has gaps: 68.5% of 2023 hours carry a real snapshot, against 96% in 2024 and essentially all hours since. Forward-filled prices would create spurious zero returns and catch-up jumps, so each bar takes the last *real* price within it, and returns that span a bar without a snapshot are dropped (146 of about

8,100 four-hourly returns, none of the daily ones).

Oracle noise and the sampling horizon. Hourly returns of the oracle-implied spread have lag-one autocorrelation of -0.16 to -0.25 depending on the period, the signature of microstructure noise (Aït-Sahalia et al., 2005), which inflates hourly volatility by up to about one half. This is why the cross-section of Appendix D.2 uses daily returns, and why the calibrations stop at four hours. Even at four hours some of it may remain: the headline model's total spread volatility (0.335) exceeds the daily realised value over the same window (0.238), so part of the fitted jump variance may be noise or short-horizon mean reversion, and part reflects the independent-jump restriction discussed in Section 8.4. Relative to four-hourly returns, the variance ratios (Lo and MacKinlay, 1988) over the full sample are 0.86 at one day, 0.88 at one week and 0.92 at one month, close to the independent increments the Lévy model assumes. They vary by period, however: 0.33 at one month in 2023, 1.56 in 2025, and 0.54 at one week in the headline window, which is unusually mean-reverting.

Full-history calibrations. Table 10 compares the headline fit with the same pipeline applied to daily bars over the full history and to four-hourly bars from 2024. Both longer samples give a larger total spread volatility. The daily fit trades many small jumps for fewer, larger ones ($\lambda \approx 85$ per year, $\eta_{2,+} \approx 3.2\%$), which is what raises the overshoot correction to -11% and the bad-debt probability to 3×10^{-4} (Section 7.4). The ECF drift normalizer $\mu_1 - \mu_2$ ranges from -0.86 to $+0.43$ across the three; it is fitted to de-meaned returns and is not an estimate of the sample drift, which Section 8 uses instead.

Calibration	σ_Z	model $\bar{\sigma}_Z$	ρ	λ_1	λ_2	$\eta_{2,+}$	ECF $\mu_1 - \mu_2$
Headline, 4h, 0.38 y	0.175	0.335	0.847	358	451	0.011	-0.86
Full history, daily, 3.7 y	0.196	0.478	0.884	90	81	0.032	-0.37
2024 onward, 4h, 2.8 y	0.235	0.485	0.794	397	277	0.015	$+0.43$

Table 10: WBTC/WETH calibrations with the headline pipeline (EWM de-meaning, penalised ECF, APR drift adjustment). σ_Z: diffusive spread volatility; model $\bar{\sigma}_Z$: total including jumps. Script `jobs/horizon_analysis.py`.

Liquidation probabilities, in and out of sample. The headline window was calm for the spread: rolling one-month windows in it crossed -0.07 only 2% of the time, while from 2024 onward they did so 36% of the time (47%, 23% and 12% for $h_0 = 0.05$, 0.10 and 0.15). Over 2024 onward the three calibrations match these frequencies to within about a factor of two, in either direction. The decisive test is out of sample (Table 11). For each of 33 months from 2024-01, the model is calibrated only on data before the month, predicts the probability that the spread falls by h_0 within it, and is scored against the realised four-hourly path. The model monitors the barrier continuously and the data only every four hours, which by itself makes the model's probabilities higher than the observed frequencies; part of the overprediction below is this monitoring effect. Three findings follow. The map's probabilities are on average *higher* than the realised frequency, by a factor of about 1.3–1.6 (two to three at $h_0 = 0.15$). Treating the 33 months as independent, the over-prediction is 1.0 to 2.8 standard errors depending on predictor and buffer. A moving-block bootstrap over months, which respects their serial dependence, gives 95% intervals for the bias that exclude zero for the four-hourly carry-only predictor at every buffer, but not for the daily fitted-drift one. The bias is therefore systematic for some predictors and not established for others. The four-hourly monitoring pushes in this direction, and so does the mean reversion of 2024 (one-month variance ratio 0.49); 2025 trended instead (1.56). A third cause is structural: with independent jumps, the calibrated model understates the legs' covariance and so overstates the spread

variance, by about 60% over the full history (Section 8.4). Daily calibration does better than four-hourly. And no predictor beats the constant base rate in Brier score, so beyond its upward bias the map has no month-to-month skill: a driftless GBM with trailing volatility does at least as well as Kou, and the fitted drift worsens the Brier score of every Kou forecast.

	mean one-month crossing probability			Brier score
Predictor	$h_0 = 0.05$	$h_0 = 0.10$	$h_0 = 0.15$	$h_0 = 0.10$
Realised frequency	0.455	0.273	0.091	0.198 (base rate)
Kou, daily, 12 months, fitted drift	0.595	0.357	0.203	0.267
Kou, daily, 12 months, carry-only drift	0.620	0.375	0.207	0.219
Kou, 4h, 6 months, fitted drift	0.585	0.386	0.258	0.323
Kou, 4h, 6 months, carry-only drift	0.678	0.436	0.258	0.226
GBM, trailing 30-day volatility	0.627	0.354	0.184	0.213

Table 11: Out-of-sample backtest of one-month liquidation probabilities, WBTC/WETH, 33 months 2024-01 to 2026-09. Each month's predictor uses only earlier data, with the headline pipeline but 8 optimiser starts instead of 30; the model horizon is $1/12$ year and the realised window 30 days. "Carry-only drift" sets the relative price drift to zero and keeps the window's net carry. The base-rate Brier score uses the realised frequency as a constant forecast. Script `jobs/rolling_backtest.py`.

Appendix D: Interest-Rate Carry and Regime Switching

On Aave the collateral earns a supply rate and the debt accrues a borrow rate, both compounding through the protocol's indices. This appendix shows that deterministic carry is a per-leg drift shift, that it is small in barrier terms for cross-asset pairs, and that the one non-level channel — the kink in the rate curve — is also small at a monthly horizon.

D.1. Carry as an effective drift

Let r_s and r_b be the supply rate on S_1 and the borrow rate on S_2, constant over $[0,T]$. Then $V_{1,t} = n_1 S_{1,0} e^{X_{1,t}+r_s t}$, $V_{2,t} = n_2 S_{2,0} e^{X_{2,t}+r_b t}$, and $h_t = h_0 + Z_t + ct$ with net carry $c := r_s - r_b$. The carry-adjusted log-prices $\widetilde{X}_{i,t} := X_{i,t} + r_i t$ ($r_1 := r_s$, $r_2 := r_b$) are Kou processes with the same diffusion and jump law; under the price-growth convention the shift is exactly $\mu_i \mapsto \mu_i + r_i$, leaving χ_i and $k\eta_{2,+} < 1$ untouched.

Proposition D.1 (Carry is an effective drift)**.** *The carry-adjusted book coincides with the book of Sections 2–5 under $(\mu_1,\mu_2) \mapsto (\mu_1 + r_s, \mu_2 + r_b)$. Survival and every killed moment $\mathbb{E}[\widetilde{\Pi}_T^k]$, $k \in \mathscr{K}$, are given by the formulas of Sections 4–5 at the shifted drifts.*

The first-passage problem is that of $\widetilde{Z} = Z + ct$ below $-h_0$, and $\widetilde{\Pi}_T$ is (3) with X_i replaced by $\widetilde{X}_i$. Positive c raises survival at fixed h_0; whether by an amount that matters is a quantitative question.

D.2. The cross-section: when deterministic carry matters

We screen the Aave v3 Ethereum universe by a rule fixed in advance — blue-chip majors and canonical wrappers, not frozen or paused, listed for the full window, ranked by USD TVL *at the window start*, collateral legs with positive origination LTV, debt legs with borrowing enabled — and report *every* admissible ordered pair: eight reserves, 37 pairs (the full table is in

`results/pair_cross_section_daily.csv`). The diagnostic is the number of barrier standard deviations that carry is worth over the horizon,

$$\mathrm{CB} := \frac{cT}{\bar{\sigma}_Z\sqrt{T}}, \tag{58}$$

where $\bar{\sigma}_Z$ is the *realised* total (diffusion plus jump) volatility of Z, measured on daily returns; CB is independent of h_0. We fix $T = 1/12$, long enough for carry to accrue yet short enough for the static assumption of Remark 2. The sampling interval matters. Oracle-implied prices carry microstructure noise at hourly frequency (Appendix C), and for near-pegged pairs that noise is most of the hourly variance: hourly sampling puts $\bar{\sigma}_Z$ for wstETH/WETH at 0.39, daily at 0.08.

Collateral	Debt	$\bar{\sigma}_Z$	$r_s - r_b$	CB	debt $U > U_{\text{opt}}$
Stablecoin pairs					
USDT	USDC	0.0039	-1.42%	-1.043	10.4%
USDC	USDT	0.0039	-0.43%	-0.318	8.5%
USDT	USDe	$<10^{-4}$	-0.51%	—	5.2%
Same-underlying pairs					
wstETH	WETH	0.078	-2.51%	-0.093	13.6%
weETH	WETH	0.176	-2.51%	-0.041	13.6%
WBTC	cbBTC	0.073	-0.30%	-0.012	0.0%
Cross-asset pairs					
WETH	USDC	0.514	-2.67%	-0.015	10.4%
WETH	WBTC	0.238	$+1.52\%$	$+0.018$	0.0%
WBTC	WETH	0.238	-2.51%	-0.030	13.6%
cbBTC	USDC	0.375	-4.54%	-0.035	10.4%

Table 12: Representative rows of the 37-pair cross-section, 2026-03-17 to 2026-08-04, $T = 1/12$. $\bar{\sigma}_Z$: annualised realised volatility of the daily log price ratio; CB: carry-to-barrier ratio (58); last column: share of hours the debt reserve's curve-implied utilization exceeds its kink (Section D.3). The USDT/USDe oracle ratio does not move at daily resolution, so CB is undefined.

Three groups emerge (Table 12). *For every cross-asset pair, deterministic carry is negligible*: $|\mathrm{CB}| \leq 0.035$, a few percent of one barrier standard deviation over a month, so it cannot change a sizing decision. *Same-underlying pairs* (a liquid-staking token against WETH, or two BTC wrappers) are near-pegged, with $\bar{\sigma}_Z$ between 0.07 and 0.18. There carry is two to three times more material, up to $\mathrm{CB} = -0.09$ for wstETH/WETH, though still under a tenth of a barrier standard deviation; their liquidation risk lies mainly in depegs (Section D.4). *Stablecoin pairs* collapse $\bar{\sigma}_Z$ to below 0.004, and there carry reaches a full barrier standard deviation ($\mathrm{CB} = -1.04$ for USDT/USDC). The barrier then sits tens of standard deviations away, so liquidation risk is carried by jumps and depegs; for USDT/USDe the oracle ratio does not move at all at daily resolution, and the barrier cannot be reached by price.

A second regularity concerns sign: *a leveraged long-crypto book generally pays carry*. Every pair with crypto collateral and stablecoin or ETH debt has net carry between -1.8% and -4.5%. Wrapped-BTC debt is the exception: those reserves sit at utilization 0.03–0.05 and charge a near-floor rate, so the carry is near zero for staking-token collateral and positive for WETH collateral (WETH/WBTC $+1.52\%$, WETH/cbBTC $+1.56\%$); even there $\mathrm{CB} \approx +0.02$.

D.3. Regime-switching carry and the 6×6 *resolvent*

With the rate *level* ruled out, what remains is its *shape*. Aave's variable borrow rate is a deterministic, piecewise-linear function of reserve utilization U,

$$r_b(U) = \begin{cases} r_0 + s_1\, U/U_{\text{opt}}, & U \le U_{\text{opt}}, \\ r_0 + s_1 + s_2\,(U - U_{\text{opt}})/(1 - U_{\text{opt}}), & U > U_{\text{opt}}, \end{cases} \tag{59}$$

with parameters published on-chain. Two consequences follow. First, the curve is invertible, so the observed rate is a sufficient statistic for utilization. This matters because the naive borrowed/supplied ratio is badly biased: for USDC it reports mean utilization 0.881 and a maximum above one, against curve-implied 0.677 and 0.931, and so misclassifies 25.4% of hours as stressed against a true 10.4%. Second, the slope jumps at U_{opt} (published s_2 of 0.60–0.80 for WETH and the major stablecoins and 3.00 for wrapped BTC, against s_1 of 0.027–0.055), so the borrower is short a call on utilization. Fitting a smooth affine diffusion to this kinked transform of a bounded state manufactures spurious jumps and mean reversion.

We therefore let net carry switch between a normal and a stressed state following a two-state Markov chain, with rates (q_1, q_2) and state carries (c_1, c_2) calibrated by thresholding utilization at U_{opt}. Only the drift switches, so this is Proposition D.1 applied state by state. The barrier system grows from 3×3 to 6×6 and reduces to two decoupled single-regime copies at zero switching (Appendix D.5, Proposition D.2).

Thresholding the WETH debt leg's hourly utilization at $U_{\text{opt}} = 0.90$ gives a stressed fraction of 19.9% and mean stressed spells of 134 hours ($q_1 = 16.3$, $q_2 = 65.5$ per year), with mean borrow rates of 2.1% (normal) and 4.0% (stressed), so $c_1 = -2.1\%$ and $c_2 = -4.0\%$. This uses the raw series; the curve-implied measure gives 13.6% stressed and would shrink the effect further. Starting from price-only drifts and adding carry back (Table 13), the deterministic mean carry $c = -2.5\%$ lowers survival by 0.55 percentage points near the barrier. The opening regime adds at most 0.09 point, and averaging the two opening states with their stationary weights reproduces the deterministic mean carry to within 0.001 point. The regime channel is *inert at a monthly horizon*: stressed spells of a few days wash out over a month. The 0.55-point cost of mean carry, which CB ≈ -0.03 does not reveal, also shows that CB understates carry's influence near the barrier, where survival is most sensitive to drift. The mechanism is not wrong, only small here: WETH and the stablecoins spend 5–14% of hours above their kinks and would show more at longer horizons or with steeper stressed rates, while wrapped-BTC debt (utilization 0.03–0.05 against $U_{\text{opt}} = 0.45$) has no rate channel at all.

h_0	price-only	det. mean carry	open normal	open stressed
0.10	0.7600	0.7545	0.7546	0.7537
0.25	0.9911	0.9906	0.9906	0.9906
0.50	1.0000	1.0000	1.0000	1.0000

Table 13: Survival at calibrated regime parameters ($q_1 = 16.3$, $q_2 = 65.5$, $c_1 = -2.1\%$, $c_2 = -4.0\%$), benchmark WBTC/WETH dynamics, $T = 1/12$. All runs start from price-only drifts (APRs of Table 2 removed). "Det. mean carry" applies the stationary $c = -2.5\%$ (Proposition D.1); "open normal/stressed" start the chain in each state. Script `regime_calibration.py`.

Remark 8 (The kink as a convexity). With $r_b(U)$ convex and U random, Jensen's inequality gives

$$\mathscr{J} \;=\; \mathbb{E}\big[r_b(U)\big] - r_b(\mathbb{E}[U]) \;>\; 0 \tag{60}$$

whenever U has mass on both sides of U_{opt}. A deterministic carry evaluated at the mean utilization would set $\mathscr{J} = 0$. The carry used here is the realised average rate, which averages r_b over the observed utilization path and therefore already contains $\mathscr{J}$. What no deterministic carry can represent is the dependence on the state in which the position opens and on when stress arrives; that is what the regime model adds, and Table 13 shows it is small at a monthly horizon.

D.4. Collateral-depeg sensitivity

A depeg that haircuts collateral by a fraction d scales H_0 by $(1-d)$, so it is the same book at cushion $h_0 + \log(1-d)$, re-evaluated on the existing map (7). It matters most for the stablecoin and same-underlying pairs the cross-section flags. Their low $\bar{\sigma}_Z$ makes a thin buffer look safe, and that apparent safety is what permits high leverage. A haircut of a few percent is many times the spread's one-month standard deviation for stablecoin pairs ($\bar{\sigma}_Z\sqrt{T} \approx 0.1\%$) and comparable to it for same-underlying pairs (2–5%), so the leverage that low $\bar{\sigma}_Z$ makes affordable is exactly what makes the book depeg-fragile. We treat the depeg as a scenario overlay so the base dynamics stay pure price. A full treatment would let it co-move with the regime channel, since the deleveraging that breaks a peg also drives utilization through the kink.

D.5. The 6×6 *regime resolvent*

Let $J_t \in \{1,2\}$ (normal, stressed) be a Markov chain with generator $Q = \left(\begin{smallmatrix} -q_1 & q_1 \\ q_2 & -q_2 \end{smallmatrix}\right)$ and net carry c_i in state i. Only the drift switches, so the exponent of Z in state i is

$$\psi^i(s) = \psi_Z^{(k)}(s) + c_i s, \tag{61}$$

with diffusion, jump law, phase rates and admissibility bounds unchanged. The killed resolvents $\widehat{U}^i$ started in state i satisfy

$$(q+q_1-\mathscr{L}^1)\widehat{U}^1 - q_1\widehat{U}^2 = g_k, \qquad -q_2\widehat{U}^1 + (q+q_2-\mathscr{L}^2)\widehat{U}^2 = g_k, \tag{62}$$

and an exponential mode $B\,v(\gamma)e^{\gamma x}$ solves the homogeneous system when v spans the kernel of $A(\gamma) = \left(\begin{smallmatrix} q+q_1-\psi^1(\gamma) & -q_1 \\ -q_2 & q+q_2-\psi^2(\gamma) \end{smallmatrix}\right)$, i.e. when $\Delta(\gamma) := \det A(\gamma) = 0$.

Proposition D.2 (Regime-switching killed resolvent)**.** *Both ψ^i share the four poles of $\psi_Z^{(k)}$; writing $\psi^i = N^i/D$ with $\deg D = 4$, $\deg N^i = 6$,*

$$\Delta(s)\,D(s)^2 = \widetilde{N}^1(s)\widetilde{N}^2(s) - q_1q_2D(s)^2, \qquad \widetilde{N}^i := (q+q_i)D - N^i,$$

a degree-12 polynomial with, generically, six roots of negative real part. The boundary conditions also number six — absorption and the two downward-phase conditions of Proposition 5.1 in each state (the coupling terms carry no generator and add no zero-extension residual) — so $B \in \mathbb{C}^6$ solves, for $i = 1,2$ and $d = 1,2$,

$$\sum_{m=1}^{6} B_m v^i(\gamma_m) = -\sum_{j=0}^{k} P_j^i e^{-jh_0}, \qquad \sum_{m=1}^{6} B_m v^i(\gamma_m)\frac{r_{d,-}}{r_{d,-}+\gamma_m} = -\sum_{j=0}^{k} P_j^i \frac{r_{d,-}}{r_{d,-}+j} e^{-jh_0},$$

where $A(j)P_j = (c_{k,j}, c_{k,j})^\top$. At $q_1 = q_2 = 0$ this is two decoupled copies of Proposition 5.1.

The kernel vector $v(\gamma)$ should be taken from the adjugate of $A(\gamma)$: in the decoupled limit one adjugate column vanishes while the other selects the state a root belongs to. The non-resonance condition becomes $\Delta(j) \neq 0$.

Lemma D.3 (Generic invertibility)**.** *With the decaying roots labelled analytically in* (q_1, q_2) *(possible while they remain simple),* $\det \mathbf{M}_{\text{bar}}$ *is analytic in* (q_1, q_2) *and at* $q_1 = q_2 = 0$ *factorises into two single-regime determinants, each non-zero by Lemma 5.2; hence the* 6×6 *system is invertible off a null set of switching rates.*

We treat invertibility at the calibrated (q_1, q_2) as numerically verified: the condition number rises only from 81 to 844 for real $q \in [0.1, 100]$ (we did not tabulate it on the complex Talbot contour). The solver reproduces the single-regime answer to 10^{-14} when $c_1 = c_2$ with $q_1 \neq q_2$ and to 10^{-15} in the decoupled limit ($k = 0, 1, 2$), and agrees with a switching Monte Carlo once its discretisation bias is refined away.

Appendix E: Proofs

Proof of Lemma 3.1. Step 1: Laplace transform under $\mathbb{P}^{(k)}$. Because $Z_t = X_{1,t} - X_{2,t}$ and the density on $\mathscr{F}_t$ equals $e^{kX_{2,t} - t\Psi(0,-ik)}$,

$$\mathbb{E}_{\mathbb{P}^{(k)}}\left[e^{sZ_t}\right] = e^{-t\Psi(0,-ik)}\,\mathbb{E}_{\mathbb{P}}\left[e^{sX_{1,t} + (k-s)X_{2,t}}\right] = e^{t[\Psi(-is,-i(k-s)) - \Psi(0,-ik)]}, \tag{63}$$

where we used $\mathbb{E}_{\mathbb{P}}[e^{\alpha X_{1,t} + \beta X_{2,t}}] = e^{t\Psi(-i\alpha,-i\beta)}$. This identity is first valid for s in the common moment-generating strip. In particular, $k\eta_{2,+} < 1$ is needed at $s = 0$ to define the tilt, and evaluating the payoff forcing at $s = j$, $j = 0, \ldots, k$, additionally requires $k\eta_{1,+} < 1$. Since the Radon–Nikodým density is exponential-linear in $(X_{1,t}, X_{2,t})$, Girsanov's theorem for Lévy processes guarantees that Z remains a Lévy process under $\mathbb{P}^{(k)}$ with exponent

$$\psi_Z^{(k)}(s) = \Psi(-is, -i(k-s)) - \Psi(0, -ik). \tag{64}$$

Step 2: Expand and simplify. Substituting (15) with $(u, v) = (-is, -i(k-s))$, and noting $u^2 = -s^2$, $v^2 = -(k-s)^2$, $uv = -s(k-s)$, and $\varphi_{J_i}(-i\cdot) = M_i(\cdot)$:

$$\begin{aligned}\Psi(-is, -i(k-s)) = {} & \mu_1^X s + \mu_2^X (k-s) + \tfrac{1}{2}\sigma_1^2 s^2 + \tfrac{1}{2}\sigma_2^2 (k-s)^2 + \rho\sigma_1\sigma_2 s(k-s) \\ & + \lambda_1(M_1(s) - 1) + \lambda_2(M_2(k-s) - 1).\end{aligned} \tag{65}$$

Setting $s = 0$ in (65) gives $\Psi(0, -ik) = \mu_2^X k + \frac{1}{2}\sigma_2^2 k^2 + \lambda_2(M_2(k) - 1)$. Subtracting and collecting terms by type:

Drift. $\mu_1^X s + \mu_2^X(k-s) - \mu_2^X k = (\mu_1^X - \mu_2^X)s$.

Diffusion. Expanding $(k-s)^2 = k^2 - 2ks + s^2$ and $(k-s) = k - s$:

$$\begin{aligned}&\tfrac{1}{2}\sigma_1^2 s^2 + \tfrac{1}{2}\sigma_2^2(k-s)^2 + \rho\sigma_1\sigma_2 s(k-s) - \tfrac{1}{2}\sigma_2^2 k^2 \\ &\quad = \tfrac{1}{2}(\sigma_1^2 + \sigma_2^2 - 2\rho\sigma_1\sigma_2)s^2 - k(\sigma_2^2 - \rho\sigma_1\sigma_2)s = \tfrac{1}{2}\sigma_Z^2 s^2 - k(\sigma_2^2 - \rho\sigma_1\sigma_2)s.\end{aligned}$$

Jumps. $\lambda_1(M_1(s) - 1) + \lambda_2(M_2(k-s) - 1) - \lambda_2(M_2(k) - 1) = \lambda_1(M_1(s) - 1) + \lambda_2(M_2(k-s) - M_2(k))$.

Combining, the drift coefficient is $\mu_Z^{(k)} := (\mu_1^X - \mu_2^X) - k(\sigma_2^2 - \rho\sigma_1\sigma_2)$, which establishes (16). The eigenrelation (17) follows because for any Lévy process with exponent ψ, the generator satisfies $\mathscr{L}e^{sz} = \psi(s)\,e^{sz}$, as can be seen by differentiating $\mathbb{E}[e^{sZ_t}] = e^{t\psi(s)}$ with respect to t at $t = 0$. □

Proof of Lemma A.1. Direct integration over the finite interval gives

$$\int_0^x e^{\beta(x-y)} r e^{-ry}\,\mathrm{d}y = e^{\beta x} r \int_0^x e^{-(r+\beta)y}\,\mathrm{d}y = e^{\beta x}\frac{r}{r+\beta}\left(1 - e^{-(r+\beta)x}\right),$$

which is (49). Since the interval is finite, this calculation is valid for every $\beta \neq -r$ without a full-line integrability condition. □

Proof of Proposition 5.1. For the ansatz (24), the zero-extension identity (53) shows that the interior resolvent equation holds precisely when its two downward-phase correction coefficients vanish. Absorption adds $\widehat{U}_k(q,0) = 0$, giving the three conditions (54). As shown in Appendix A, substitution of the ansatz produces exactly the system (26) with matrix (27) and forcing vector (28). Lemma 5.2 gives the unique coefficient vector.

The resulting zero extension solves (23); uniqueness of the killed resolvent in the relevant exponential-growth class identifies it with $\widehat{U}_k$. Finally, (21) and evaluation at $x = h_0$ give (29). □

Proof of Lemma 5.2. For brevity, write $\gamma_m := \gamma_m^{(k)}(q)$. Subtracting row 1 from rows 2 and 3 replaces entry (i,m) for $i = 2,3$ by

$$-\frac{\gamma_{m+3}}{r_{i-1,-}^{(k)} + \gamma_{m+3}}.$$

Pulling out -1 from each of these two rows (net sign $(-1)^2 = 1$) and then factoring γ_{m+3} from each column gives

$$\det\big(\mathbf{M}_{\mathrm{bar}}^{(k)}\big) = \gamma_4\gamma_5\gamma_6 \, \det\left(\frac{1}{x_i + \gamma_{j+3}}\right)_{1\le i,j\le 3},$$

where $x_1 := 0$, $x_2 := r_{1,-}^{(k)}$, $x_3 := r_{2,-}^{(k)}$. The right-hand factor is a Cauchy determinant; the Cauchy formula yields

$$\det\left(\frac{1}{x_i + \gamma_{j+3}}\right) = \frac{\prod_{1\le i<i'\le 3}(x_{i'} - x_i)\cdot\prod_{4\le m<m'\le 6}(\gamma_{m'} - \gamma_m)}{\prod_{i=1}^{3}\prod_{m=4}^{6}(x_i + \gamma_m)}.$$

Since $x_1 = 0$, the factor $\prod_{m=4}^{6}(x_1 + \gamma_m) = \gamma_4\gamma_5\gamma_6$ cancels with the prefactor, yielding a ratio whose every factor is nonzero: $r_{1,-}^{(k)} > 0$ since $\eta_{1,-} > 0$; $r_{2,-}^{(k)} > 0$ since $k\eta_{2,+} < 1$; $(r_{2,-}^{(k)} - r_{1,-}^{(k)}) \neq 0$ by (25); the γ_m are pairwise distinct by Assumption 5.1(ii); and $r_{j,-}^{(k)} + \gamma_m \neq 0$ because $-r_{j,-}^{(k)}$ is a pole of $\psi_Z^{(k)}$, so $\psi_Z^{(k)}(-r_{j,-}^{(k)}) = \infty \neq q$, hence $\gamma_m \neq -r_{j,-}^{(k)}$. □

The survival probability requires no separate proof: it is the $k = 0$ instance of Proposition 5.1 and Lemma 5.2 above. At $k = 0$, the forcing term in (24) is $1/q$ because $\psi_Z^{(0)}(0) = 0$, while (25) becomes $\eta_{1,-} \neq \eta_{2,+}$.

Proof of Proposition 7.1. Let $D_t := h_0 + Z_t > 0$ be the distance to the barrier before liquidation. On $\{\tau \le T\}$, almost surely exactly one of three things happens at τ: the diffusion carries Z continuously onto the barrier ($u = 0$, possible because $\sigma_Z > 0$); X_1 jumps down; or X_2 jumps up. Simultaneous jumps have probability zero because N_1 and N_2 are independent. Down-jumps of X_1 arrive at rate $\lambda_1(1-p_1)$ with sizes exponential at rate $r_1 = 1/\eta_{1,-}$, independent of the past; a jump of size y at time $t < \tau$ liquidates exactly when $y > D_{t-}$, with overshoot $u = y - D_{t-}$. By the compensation formula for the Poisson random measure of these jumps, for bounded measurable f,

$$\mathbb{E}\big[f(u)\,\mathbf{1}_{\{\tau\le T,\ X_1 \text{ jump}\}}\big] = \mathbb{E}\int_0^{T\wedge\tau} \lambda_1(1-p_1)\int_{D_{t-}}^{\infty} f(y - D_{t-})\, r_1 e^{-r_1 y}\,\mathrm{d}y\,\mathrm{d}t = P_1\int_0^{\infty} f(u)\, r_1 e^{-r_1 u}\,\mathrm{d}u,$$

with $P_1 := \mathbb{E}\int_0^{T\wedge\tau}\lambda_1(1-p_1)e^{-r_1 D_{t-}}\mathrm{d}t$, after substituting $y = D_{t-} + u$. The inner integral factorises because the exponential density satisfies $r_1 e^{-r_1(D+u)} = e^{-r_1 D}\, r_1 e^{-r_1 u}$. The same argument with rate $r_2 = 1/\eta_{2,+}$ and intensity $\lambda_2 p_2$ applies to up-jumps of X_2, and continuous crossings contribute $P_0 f(0)$. The three cases exhaust $\{\tau \le T\}$, so $P_0 + P_1 + P_2 = p_{\mathrm{liq}}(T)$ and (38) follows. With $f(u) = e^{-ju}$ the right-hand side is $P_0 + P_1 r_1/(r_1 + j) + P_2 r_2/(r_2 + j)$. The matrix with rows $\big(1, r_1/(r_1+j), r_2/(r_2+j)\big)$, $j = 0,1,2$, has determinant proportional to $r_1 - r_2$, so it is nonsingular when $r_1 \neq r_2$ and the weights are determined. □

Replication

All results are produced by the `optimal_long_short` Python package: the resolvent and moment machinery in `moments.py`, `laplace_resolvent.py`, `root_finder.py` and `inversion.py`; exact overshoot recovery in `overshoot_recovery.py`; the overshoot law and expected log wealth in `kelly.py`; the regime resolvent in `regime_switching.py`; stale-aware full-history panels in `full_history.py`; and ECF calibration in the `calibration/` subpackage. Scripts under `jobs/`: `build_merged_4h.py` and `calibrate_btc_eth.py -merged` (headline calibration); `build_benchmark_params.py` (the benchmark of Table 2, which replaces the ECF drift normalizers by sample drifts and is written to `results/params_WBTC_WETH_benchmark.json`). The results of Sections 6–7 and Table 13 are produced by `numerical_comparison.py` (Table 3), `mc_convergence.py` (Table 4), `sensitivity_analysis.py` (Figure 1), `overshoot_table.py` (Table 5), `regime_calibration.py` (Table 13) and `kelly_sizing.py` (the validation of Section 7.5), each run with the benchmark file in place of `results/params_WBTC_WETH_headline.json`; `run_merged_benchmark.sh` does this in a temporary copy of the repository and collects the outputs in `results/merged/`. Section 8: `decision_rules.py` (Table 6), `breakeven_drift.py` (Table 7, Figure 2 and the posteriors), `realised_backtest.py` and `fig_realised_backtest.py` (Table 8, Figure 3). Appendices: `bootstrap_se.py` (Table 9), `model_comparison.py`, `fetch_universe.py`, `horizon_analysis.py`, `rolling_backtest.py` and `backtest_bias.py` (Tables 10–11). Data are the Aave v3 Ethereum series under `data/AAVE/` (headline window) and `data/AAVE_full/` (full history).

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